CS6004NI: Application Development
30% Group Coursework
Autumn 2025-26
Credit: 30

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Module Leader: Mr. Bikram Poudel

Project File Links:

Link:	https://github.com/Kushal-C-Shrestha/apd-frontend https://github.com/Kushal-C-Shrestha/apd-backend

I confirm that I understand my coursework needs to be submitted online via My Second Teacher under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a marks of zero will be awarded.

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1 Overview

This project entails the development of a complete web application designed for a vehicle service and parts retail center. The system is intended to streamline and efficiently manage day-to-day operations, including the tracking of inventory, the processing of sales, customer management, and financial record-keeping. With distinct role-based access for admins, staff and customers alike, users are restricted to the tasks for which they are authorized. Essential features of the system include part management, invoice creation, service booking and automated notification generation to improve the efficiency of the operation and minimize manual intervention.

As well as these core features, the system endeavors to offer an intelligent and user-friendly experience. Data is organized in a systematic fashion allowing immediate access to information; staff are able to quickly find part and customer information whereas admins can readily view stock levels and performance reports. The customer too can enjoy many features such as viewing their own service history and being automatically alerted when vehicle maintenance is due.

This project is aimed to mimic a real-world business environment where multiple individuals need to use the same application to complete tasks. Not only are functional requirements met, but also user-friendliness, security and scalability; a practical tool for managing a vehicle service and parts business.

1.1 Purpose

The purpose of this project is to practice and demonstrate the skills I've acquired with C# and ASP.NET Core through the development of a real-world application. In doing so, I will be able to showcase the knowledge I've gained in aspects such as; System Design, Coding practices, database implementation and user interface creation.

In addition to the practical nature of this project, I intend to create efficient and well-written code with appropriate error handling and security considerations; and present it alongside a detailed report to show how I achieved this, the features it implements and why the decisions that have been made have been made.

1.2 Scope

The system will support three user types:

Admin: Manage Staff, Venders, Vehicle Parts, purchases and produce reports. They will monitor stock and overall business performance.

Staff: Manage everyday operations of the business such as, customer registration, sale processing, stock search and send customers notifications and emails.

Customer: Be able to register / log into the application to book vehicle services and view their service history and purchases history, also to be able to receive vehicle maintenance notifications (AI suggestions possible if included).

All of the required features such as low stock alerts, loyalty discount, invoice generation, and reports will be developed. The application will be implemented using ASP.NET Core with PostgreSQL.

1.3 Aims and Objectives

1.3.1 Aim

The main objective of this project is to create a simple, user-friendly and efficient web application for the real-time handling of vehicle services, parts inventory, sales, and customer communication processes.

1.3.2 Objectives

The objectives of this project is listed below:

- To implement all the main functionalities for parts and vendors CRUD operations, sales operations, invoice generation and customer reports.
- To add extended features including automated e-mail notifications, low-stock alerts, and loyalty discounts system.
- To produce quality code that is structured well, highly readable and incorporates appropriate error handling.

- To develop an easy-to-use interface for the application that is quick and efficient for any user to navigate.
- To create proper documentation of this project.

2 Tech Stack

- Frameworks: For backend of the application, .NET 8.0 is used which as ASP.NET Core Web API serves as main framework to deal with business logic, API requests, authentication, and database manipulation. For the frontend, React is utilized in order to produce responsive user interface.

  Backend: .NET 8.0 (ASP.NET Core Web API)
  Frontend: React 19 (with Vite)

- External libraries:
  Backend: To support database interaction, authentication and API documentation, following libraries have been used.

  - Entity Framework Core (v8.0.0): The object-relational mapper used to easily translate the C# object into the corresponding database tables.

  - Npgsql.EntityFrameworkCore.PostgreSQL (v8.0.0): PostgreSQL database provider to establish communication between Entity Framework Core and the PostgreSQL database.

  - Microsoft.AspNetCore.Authentication.JwtBearer (v8.0.0): Provides mechanism to authenticate users through JSON web token, allowing access only for authorized users to protected API routes.

  - Microsoft.AspNetCore.Identity.EntityFrameworkCore (v8.0.0): This gives us basic support for user authentication and management of roles, we are able to perform role based access for Admin, Staff and Customers.

- Swashbuckle.AspNetCore (v6.6.2): To generate the Swagger UI to test API endpoints on the fly during the development phase.

Frontend : Following are the important libraries used for frontend development.

- React (v19.2.5): A JavaScript library which is used to create the User Interface of the application. Using React we can create reusable components which leads to faster, interactive, and dynamic UI.

- React DOM (v19.2.5): Used with React to render React components to the DOM in the browser. It can efficiently update and render the DOM.

- React Router DOM (v7.14.2): Used for managing routes within the application. Single-page application navigation could be handled, like navigating between different views such as login, dashboard, inventory, etc, without refreshing the whole page.

- Axios (v1.15.2): Promise-based HTTP client used to make HTTP requests from the frontend to the backend API. Primarily used to retrieve data, add new data, update and delete data from the server.

- Lucide React (v1.11.0): This is a clean and customizable icon library used for designing User Interface elements.

- Database: PostgreSQL

3 Features and functionalities

3.1 Completion status

S.No	Feature	Assigned to	Completion status
1.	Admin can generate and view financial reports (daily, monthly, yearly)	Pratik	Completed
2.	Admin can manage staff registration and roles	Bipin	Pending
3.	Admin can perform parts management (purchase, edit, delete)	Dishant	Completed
4.	Admin can create purchase invoices for stock updates	Dishant	Pending
5.	Admin can manage vendor details (CRUD operations)	Dishant	Completed
6.	Staff can register new customers with vehicle details	Bipin	Completed
7.	Staff can sell vehicle parts and create sales invoices	Slesha	Completed
8.	Staff can view customer details, history, and vehicle info	Slesha	Completed
9.	Staff can generate customer-related reports (regulars, high spenders, pending credits)	Pratik	Completed
10	Staff can search customers by vehicle number, phone, ID, or name	Bipin	Completed
11	Staff can send invoices via email to customers	Slesha	Pending
12	Customers can self-register and manage profile & vehicle details	Bipin	Pending
13	Customers can book appointments, request unavailable parts, and review services	Kushal	Completed
14	Customers can view their purchase/service history	Pratik	Pending
15	System automatically notifies Admin for low stock (<10) and sends email reminders to customers with unpaid credits for more than 1 month	Kushal	Pending
16	Loyalty Program: Customers get 10% discount if they spend more than 5000 in a single purchase	Kushal	Pending

Table 1: Feature completion status table.

3.2 Completed features

3.2.1 Member 1: Bipin Kumar Thapa

Feature 6: Staff can register new customers with vehicle details

Frontend:

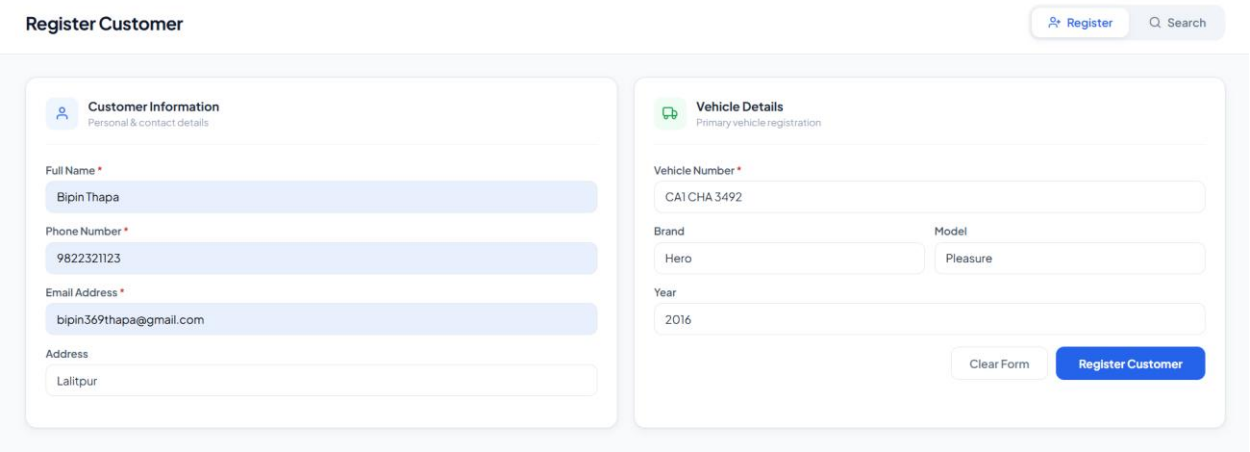


Figure 1: Adding new customer with vehicle details from Frontend

Backend:

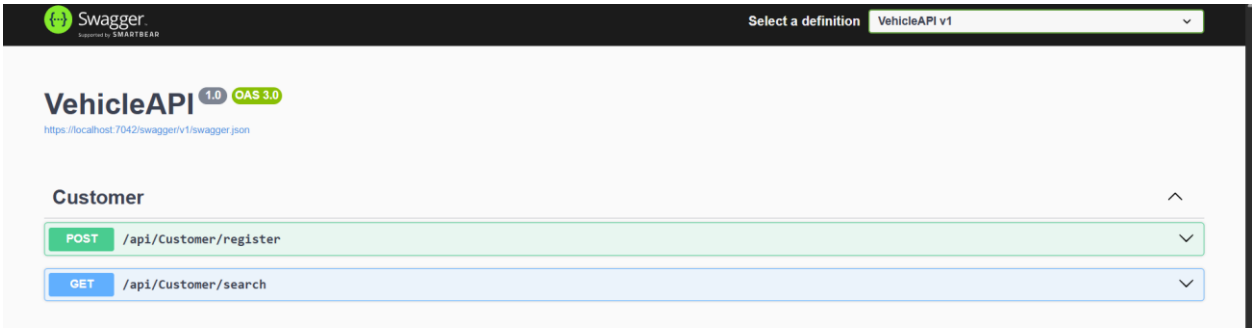


Figure 2: Backend API for customer registration and Search

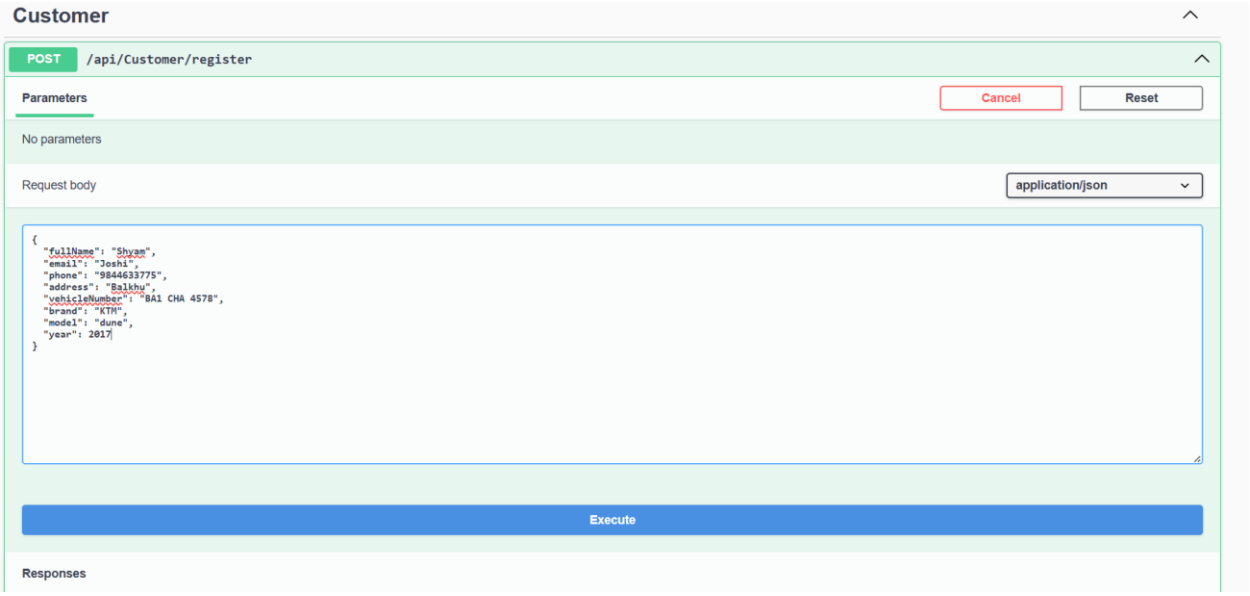


Figure 3: Details required to register

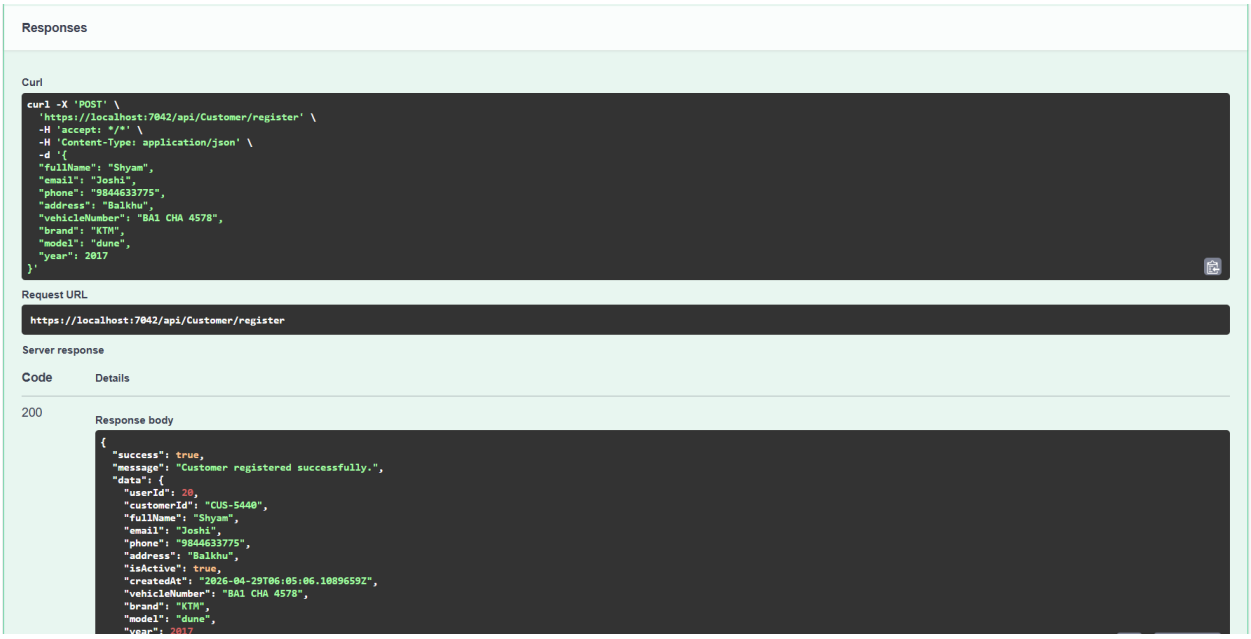


Figure 4: Successfully Customer registration.

Feature 10: Staff can search customers by vehicle number, phone, ID, or name

Frontend:

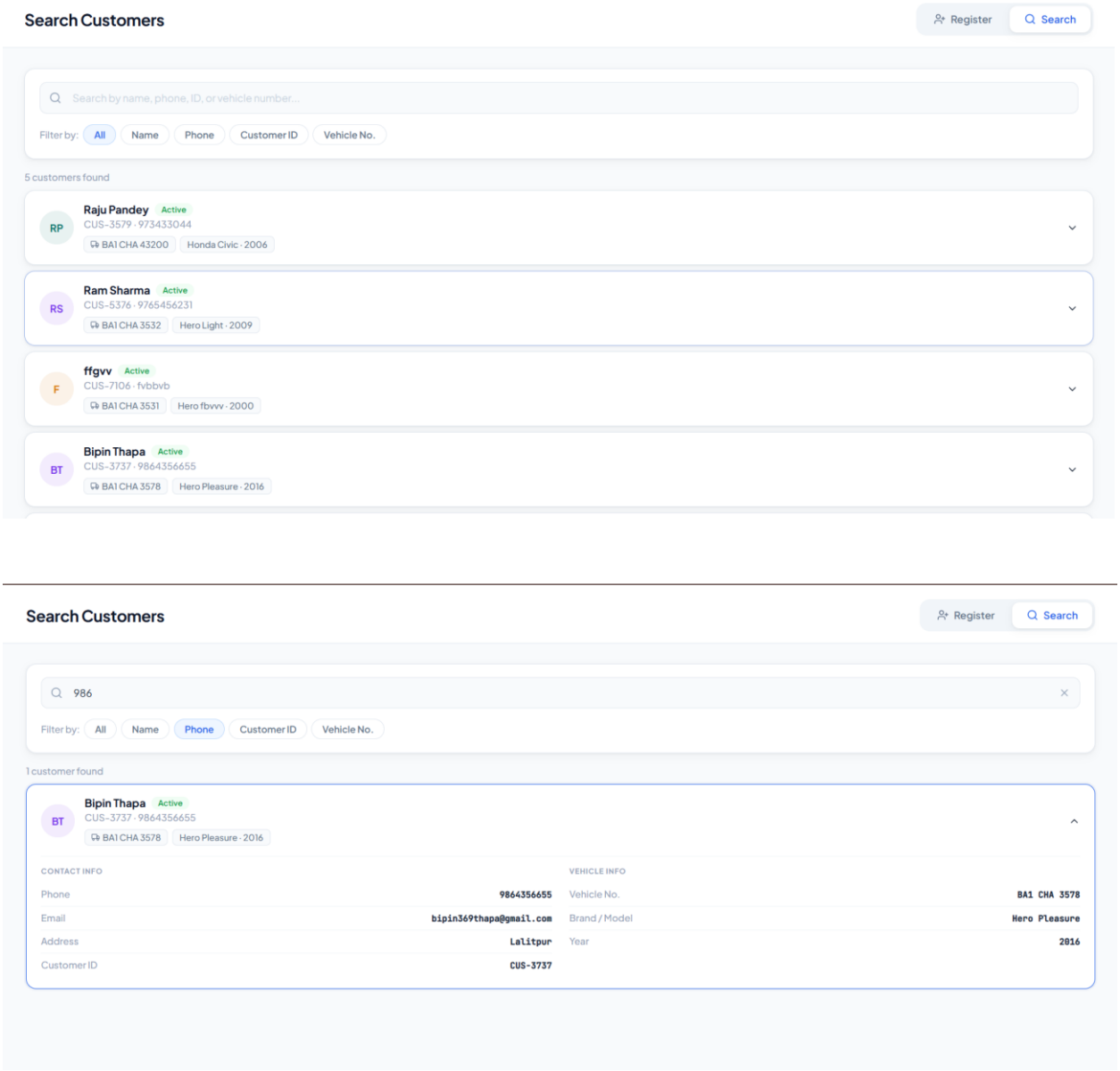


Figure 5: Customer Search Interface

Backend:

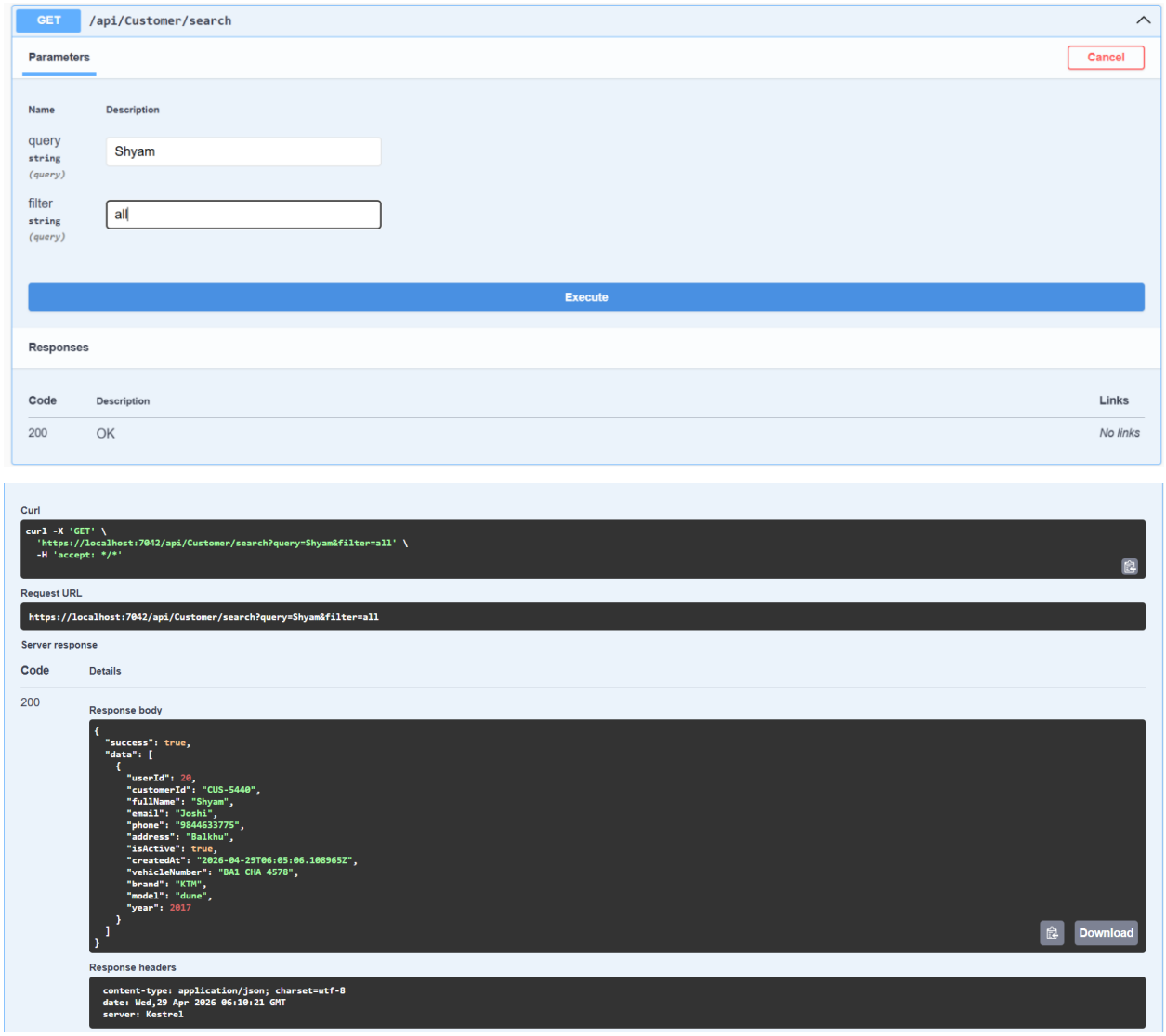


Figure 6: Customer Search API Response

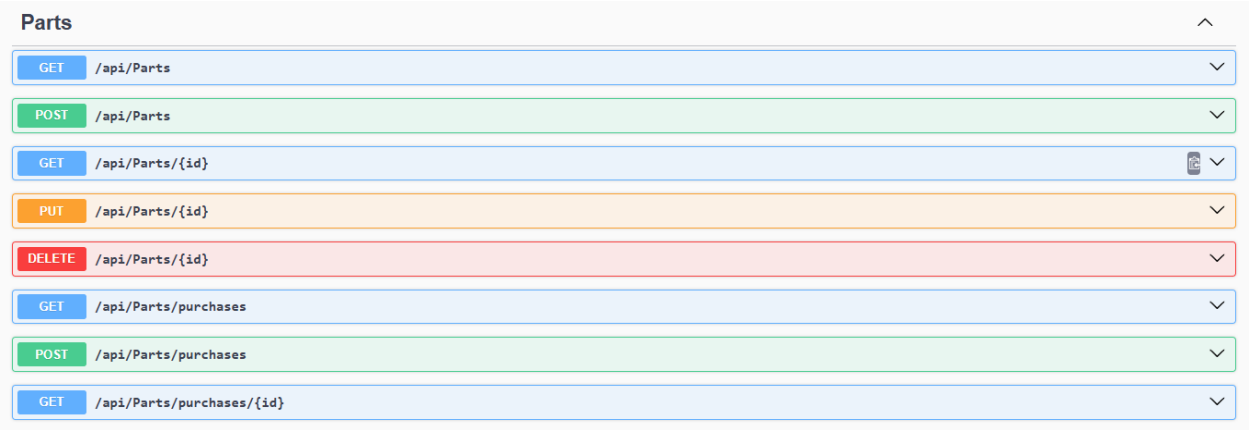
POST /api/Customer/register - Registers a new customer along with vehicle details. It automatically generates a customer ID and password after registration and sends them to the customer's email.

GET /api/Customer/search - Fetches all registered customers with detailed information. Allows filtering by name, phone, address, and vehicle number.

3.2.2 Member 2: Dishant Panjiyar

Feature 3: Admin can perform parts management (purchase, edit, delete)

Backend:



- **GET /api/Parts** - Fetches all parts from the database and returns them as a list. This is what loads Parts in Inventory table.
- **POST /api/Parts** - Creates a new part. Receives name, description, unit price, and stock quantity, saves it to the database, and returns the created part.
- **GET /api/Parts/{id}** - Fetches a single part by its ID. Used internally when needed to provide details of one specific part.
- **PUT /api/Parts/{id}** - Updates an existing part by ID. Receives the updated name, price, stock quantity etc. and overwrites the existing record.
- **DELETE /api/Parts/{id}** - Deletes a part by ID permanently from the database. Returns a conflict error if the part is referenced in purchases.
- **GET /api/Parts/purchases** - Fetches all purchase records with their items and vendor info. This is what loads the Purchase History table.
- **POST /api/Parts/purchases** - Records a new purchase. Receives a vendor ID and list of items (part, quantity, unit cost), saves the purchase, and automatically increases the stock quantity of each part involved.
- **GET /api/Parts/purchases/{id}** - Fetches a single purchase record by ID including all its line items.

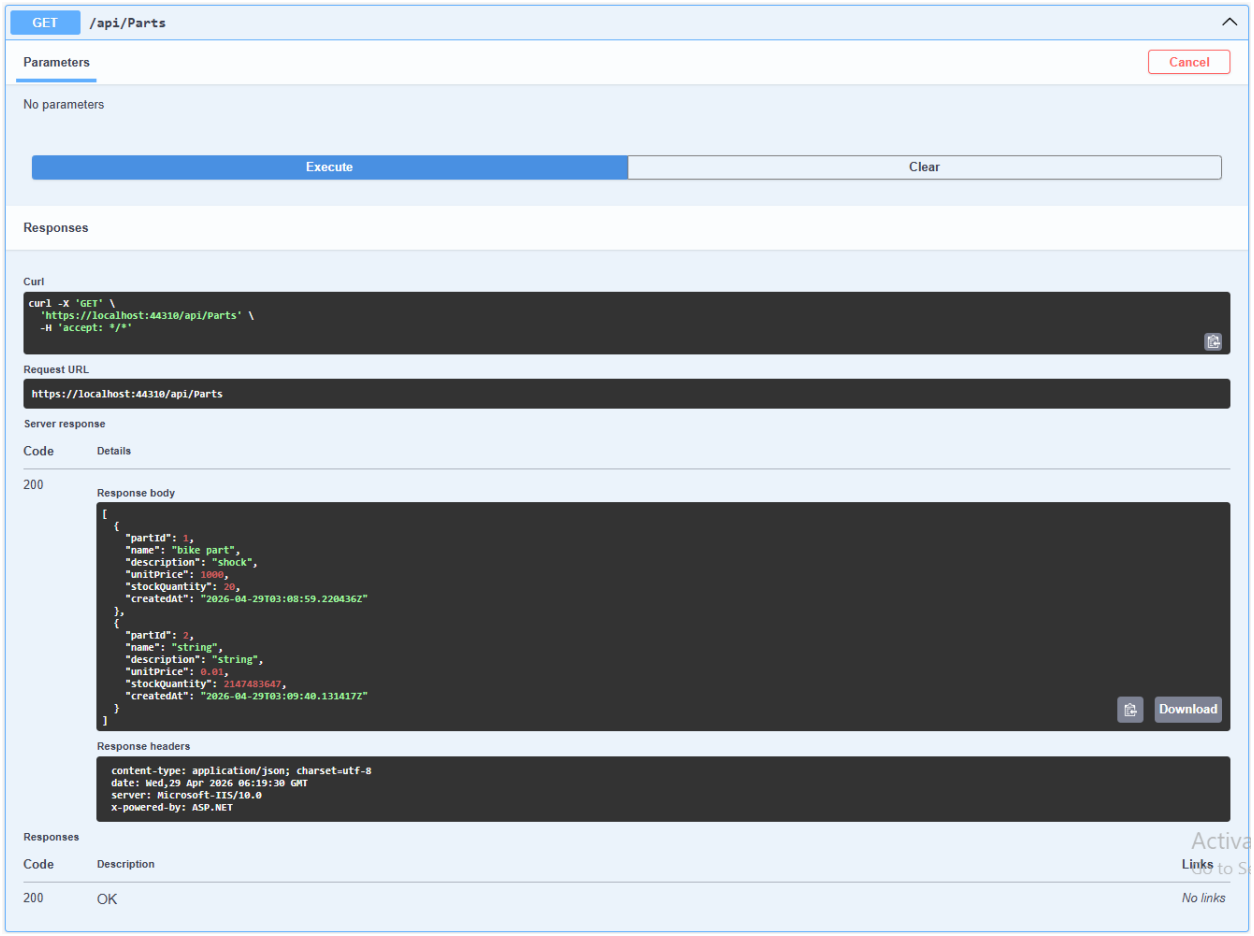


Figure 7: Parts List Retrieval API (GET /api/Parts)

Query | Query History

```
SELECT * FROM public."Parts"
ORDER BY "PartId" ASC
```

Data Output | Messages | Notifications

Showing rows: 1 to 2

	PartId [PK] integer	Name character varying (100)	Description text	UnitPrice numeric	StockQuantity integer	CreatedAt timestamp with time zone
1	1	bike part	shock	1000	20	2026-04-29 08:53:59.220436+05:...
2	2	string	string	0.01	2147483647	2026-04-29 08:54:40.131417+05:...

DELETE /api/Parts/{id}

Parameters | Cancel

Name | Description

id * required
integer($int32)
(path)
2

Execute | Clear

Responses

Curl

```
curl -X 'DELETE' \
  'https://localhost:44310/api/Parts/2' \
  -H 'accept: */*'
```

Request URL

```
https://localhost:44310/api/Parts/2
```

Server response

Code	Details
200	Response body

```
{
  "message": "Part deleted successfully."
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Wed,29 Apr 2026 06:24:13 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

Responses

Code	Description	Links
200	OK	No links

Figure 8: Delete Part Operation (DELETE /api/Parts/{id}

Query | Query History

```
1  SELECT * FROM public."Parts"
2  ORDER BY "PartId" ASC
```

Data Output | Messages | Notifications

Showing rows: 1 to 1

	PartId [PK] integer	Name character varying (100)	Description text	UnitPrice numeric	StockQuantity integer	CreatedAt timestamp with time zone
1	1	bike part	shock	1000	20	2026-04-29 08:53:59.220436+05:...

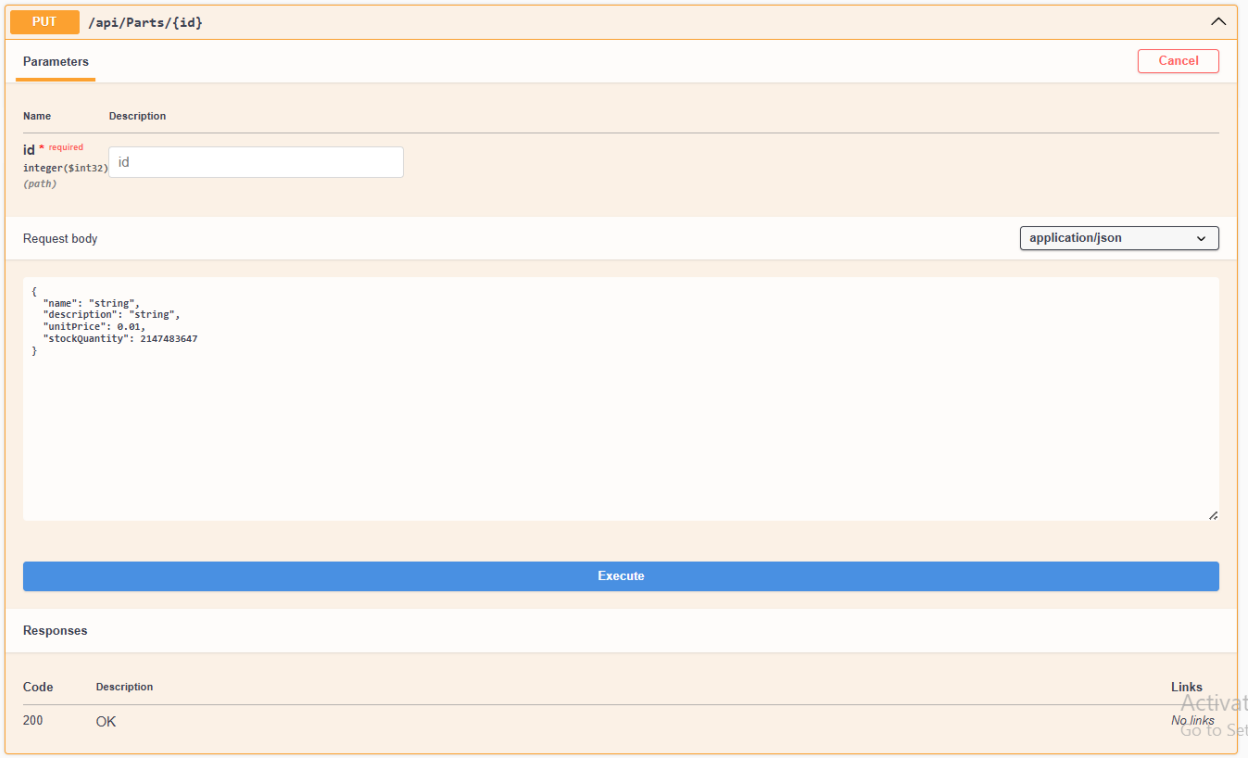


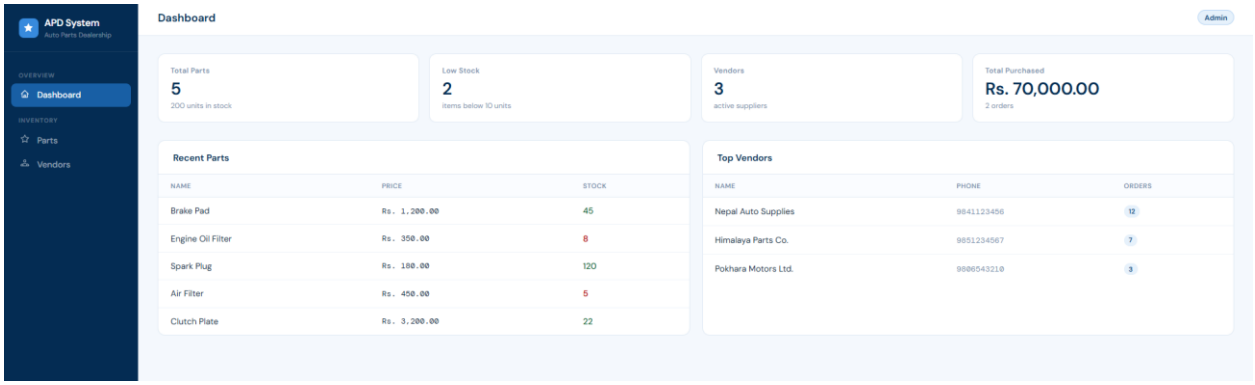
Figure 9: Update Part Details (PUT /api/Parts/{id})

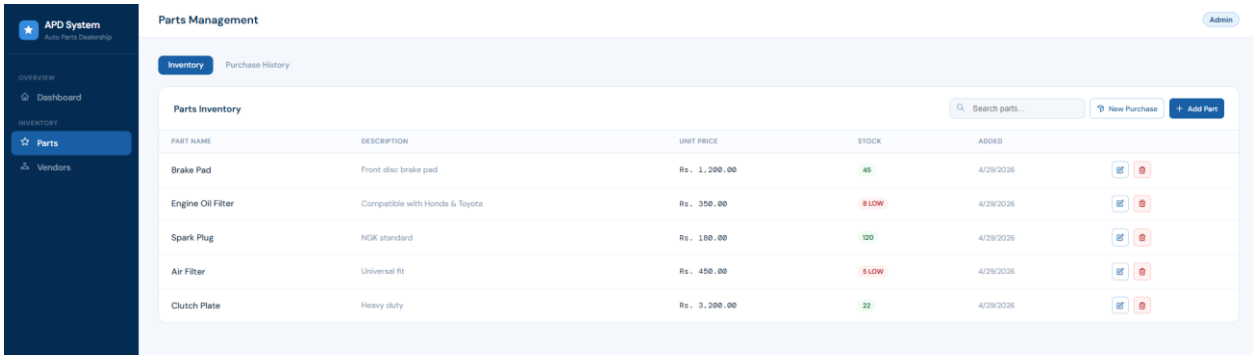
Schemas

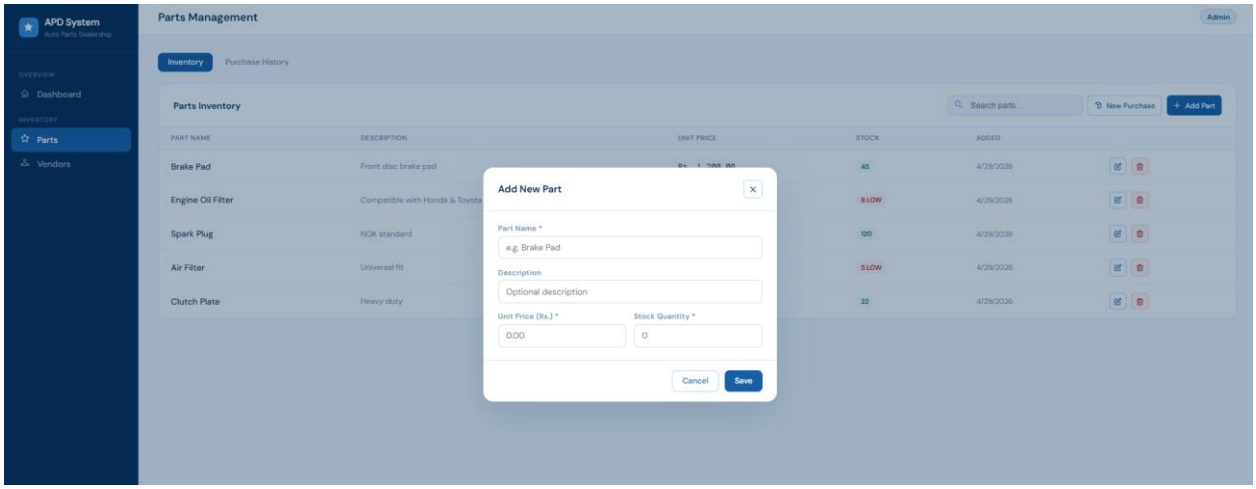
```
CreatePartDTO {
  name*           > [...]
  description     > [...]
  unitPrice*      > [...]
  stockQuantity*  > [...]
}

CreatePurchaseDTO {
  vendorId*       > [...]
  items*          > [...]
}

CreateVendorDTO {
  name*           > [...]
  phone*          > [...]
  email           > [...]
  address         > [...]
}

PurchasePartItemDTO {
  partId*         > [...]
  quantity*       > [...]
  unitCost*       > [...]
}

UpdatePartDTO {
  name*           > [...]
  description     > [...]
  unitPrice*      > [...]
  stockQuantity*  > [...]
}

UpdateVendorDTO {
  name*           > [...]
  phone*          > [...]
  email           > [...]
  address         > [...]
}
```

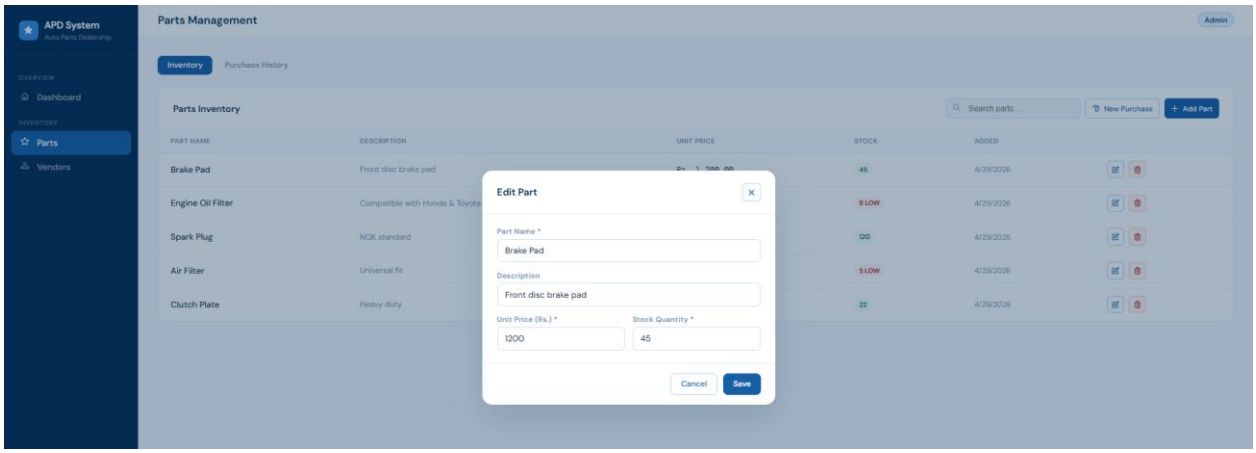
Activa
Go to Se

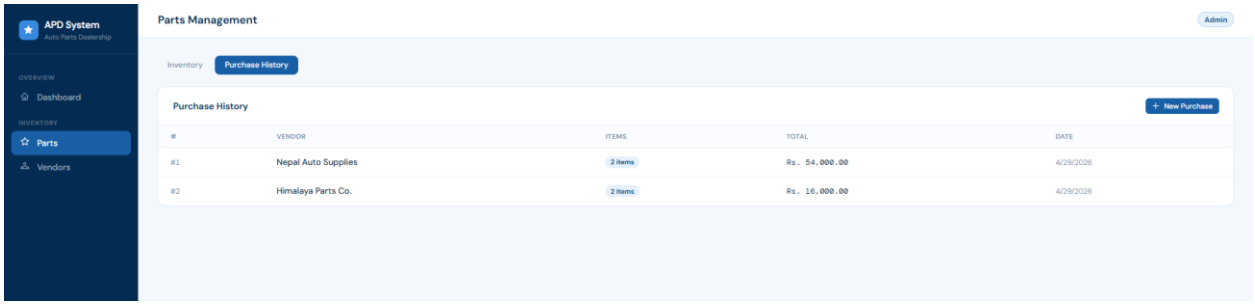
Frontend:

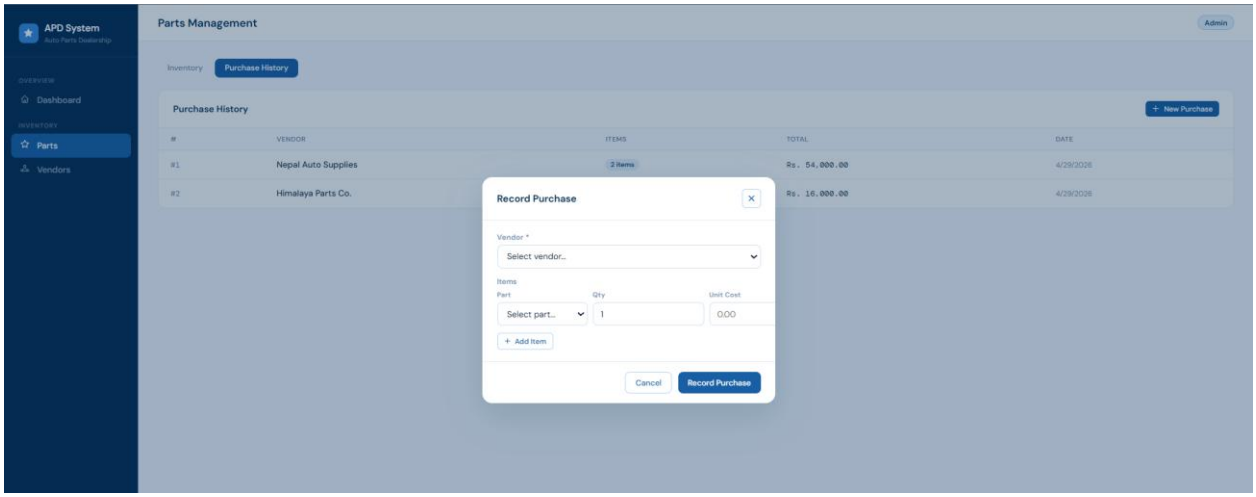





APD System
Auto Parts Dealership
Parts Management
Admin
Inventory
Purchase History
Dashboard
Parts
Vendors
Parts Inventory
Search parts...
New Purchase
Add Part
PART NAME
DESCRIPTION
UNIT PRICE
STOCK
ADDED
Brake Pad
Front disc brake pad
45
4/29/2026
Engine Oil Filter
8 LOW
Spark Plug
120
Air Filter
5 LOW
Clutch Plate
Heavy duty
22
Add New Part
Part Name *
e.g. Brake Pad
Description
Optional description
Unit Price (Rs.) *
0.00
Stock Quantity *
0
Cancel
Save

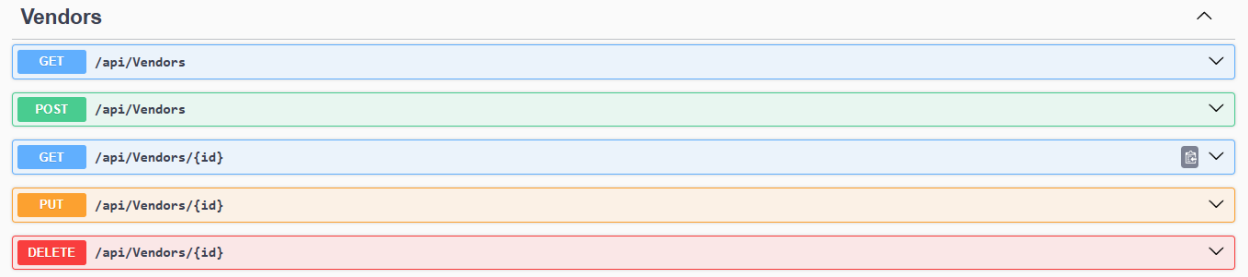

APD System
Auto Parts Dealership
Parts Management
Admin
Inventory
Purchase History
Dashboard
Parts
Vendors
Parts Inventory
Search parts...
New Purchase
Add Part
PART NAME
DESCRIPTION
UNIT PRICE
STOCK
ADDED
Brake Pad
Front disc brake pad
45
4/29/2026
Engine Oil Filter
8 LOW
Spark Plug
120
Air Filter
5 LOW
Clutch Plate
Heavy duty
22
Edit Part
Part Name *
Brake Pad
Description
Front disc brake pad
Unit Price (Rs.) *
1200
Stock Quantity *
45
Cancel
Save


APD System
Auto Parts Dealership
OVERVIEW
Dashboard
INVENTORY
Parts
Vendors
Parts Management
Admin
Inventory
Purchase History
Purchase History
+ New Purchase
# VENDOR ITEMS TOTAL DATE
#1 Nepal Auto Supplies 2 items Rs. 54,000.00 4/29/2026
#2 Himalaya Parts Co. 2 items Rs. 16,000.00 4/29/2026

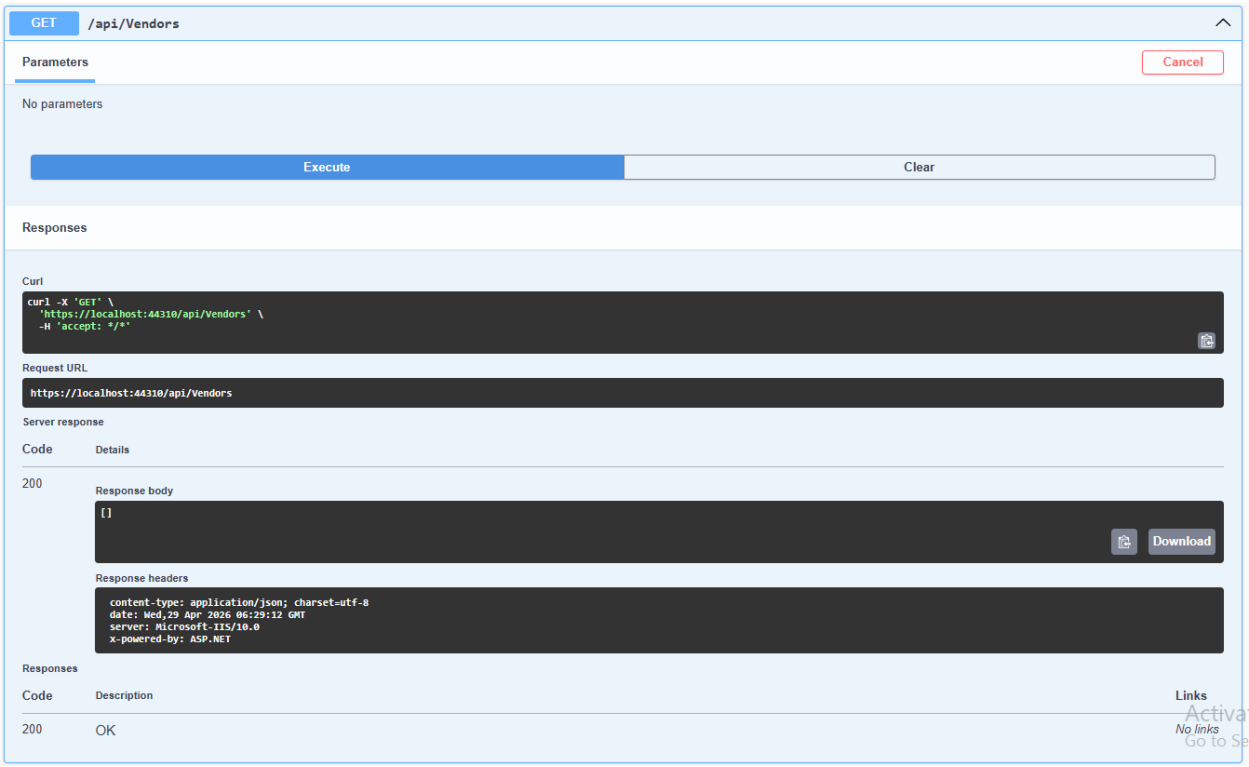

APD System
Auto Parts Dealership
Dashboard
Parts
Vendors
Parts Management
Admin
Inventory
Purchase History
Purchase History
+ New Purchase
# VENDOR ITEMS TOTAL DATE
#1 Nepal Auto Supplies 2 items Rs. 54,000.00 4/29/2026
#2 Himalaya Parts Co. Rs. 16,000.00 4/29/2026
Record Purchase
Vendor *
Select vendor...
Items
Part
Qty
Unit Cost
Select part...
1
0.00
+ Add Item
Cancel
Record Purchase

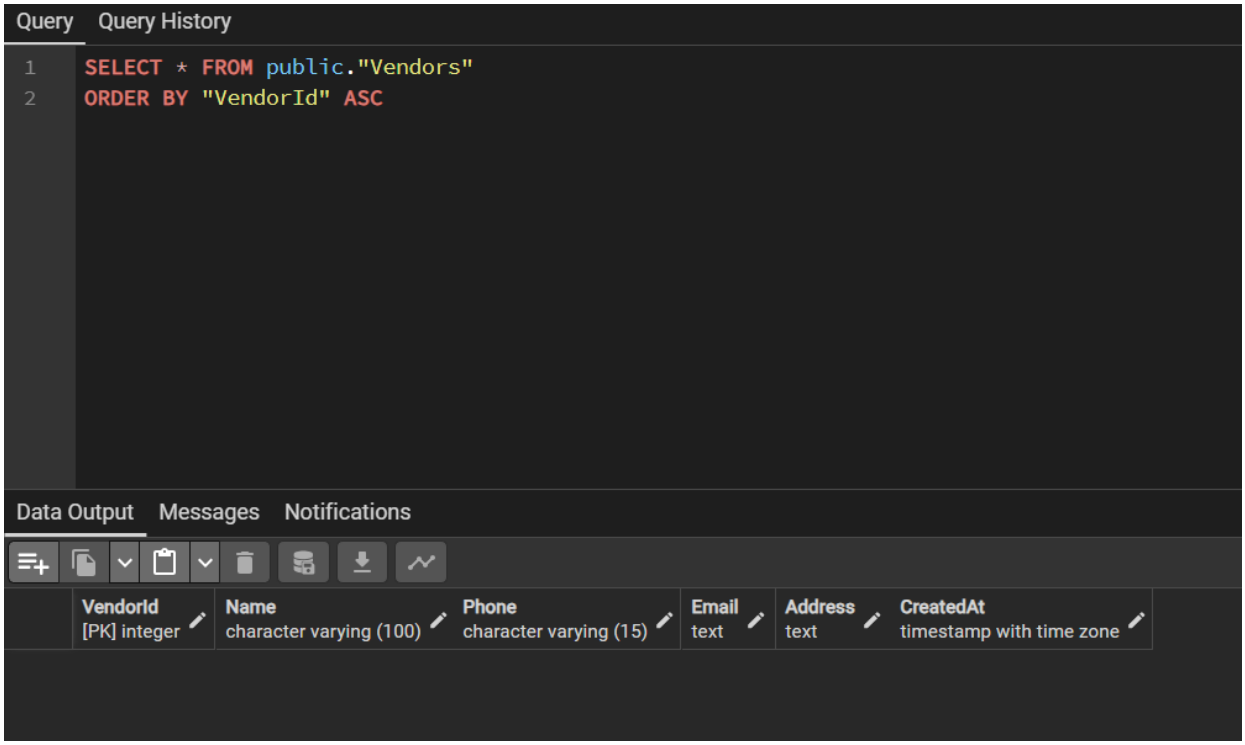
3.2.3 Feature 5: Admin can manage vendor details (CRUD operations)

Backend:



- **GET /api/Vendors** - Fetches all vendors from the database. This is what loads Vendor Directory table.
- **POST /api/Vendors** - Creates a new vendor. Receives name, phone, email, and address and saves it to the database.
- **GET /api/Vendors/{id}** - Fetches a single vendor by ID including their total purchase count.
- **PUT /api/Vendors/{id}** - Updates an existing vendor's details by ID such as phone number or address.
- **DELETE /api/Vendors/{id}** - Deletes a vendor by ID. Returns a conflict error if the vendor has existing purchases linked to them.


GET
/api/Vendors
Parameters
Cancel
No parameters
Execute
Clear
Responses
Curl
curl -X 'GET' \
'https://localhost:44310/api/Vendors' \
-H 'accept: */*'
Request URL
https://localhost:44310/api/Vendors
Server response
Code
Details
200
Response body
[]
Download
Response headers
content-type: application/json; charset=utf-8
date: Wed,29 Apr 2026 06:29:12 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
Responses
Code
Description
Links
200
OK
No links


Query
Query History
SELECT * FROM public."Vendors"
ORDER BY "VendorId" ASC
Data Output
Messages
Notifications
VendorId
[PK] integer
Name
character varying (100)
Phone
character varying (15)
Email
text
Address
text
CreatedAt
timestamp with time zone

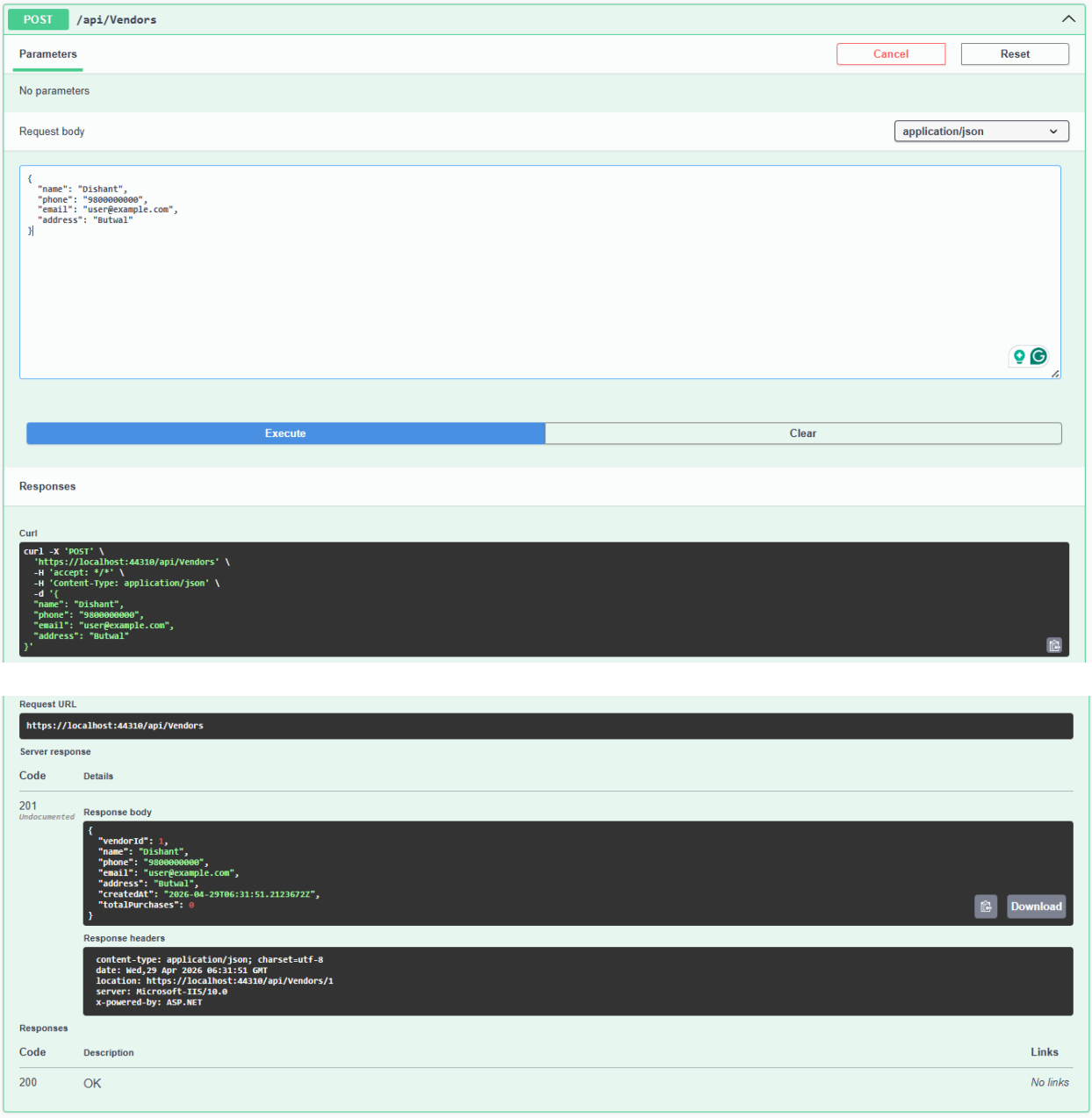


Figure 10: Add New Vendor

Query Query History

```
1  SELECT * FROM public."Vendors"
2  ORDER BY "VendorId" ASC
```

Data Output Messages Notifications

Showing rows: 1 to 1

	VendorId [PK] integer	Name character varying (100)	Phone character varying (15)	Email text	Address text	CreatedAt timestamp with time zone
1	1	Dishant	9800000000	user@example.co...	Butwal	2026-04-29 12:16:51.212367+05:...

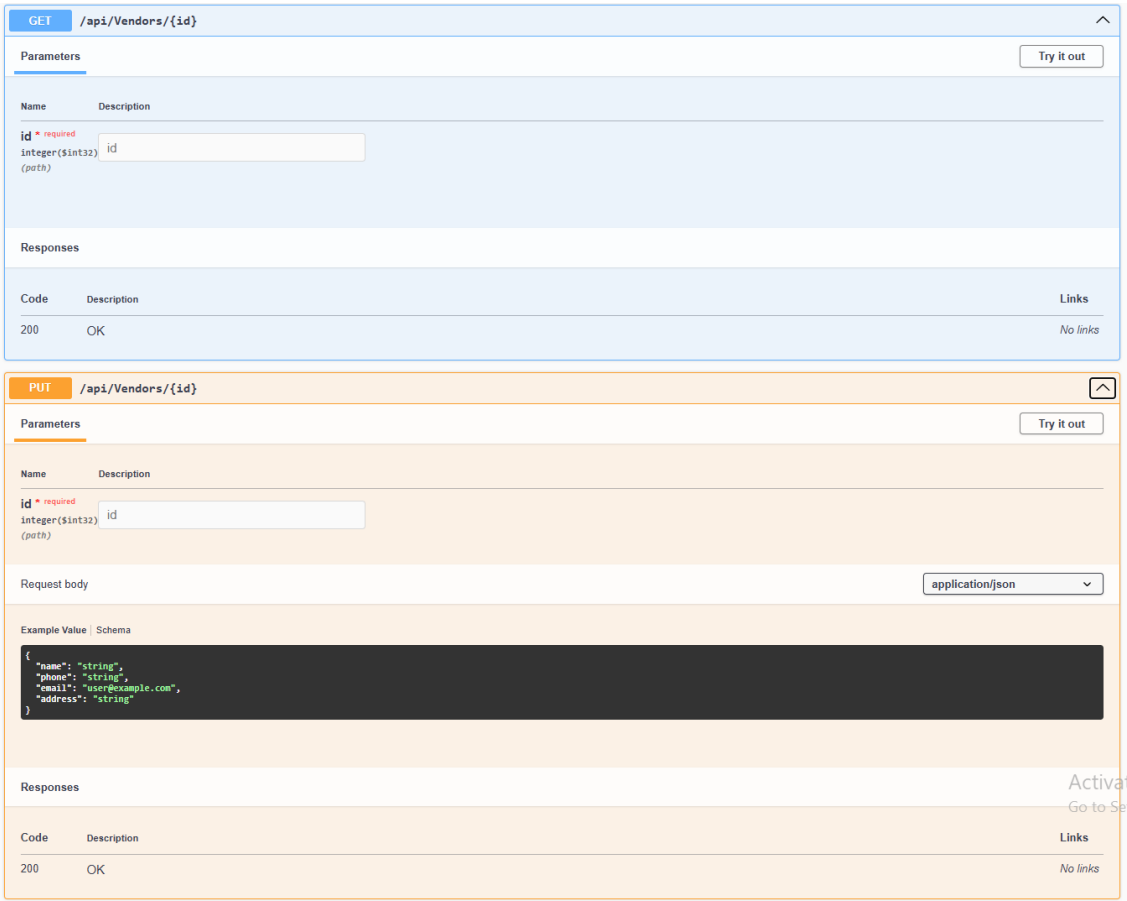
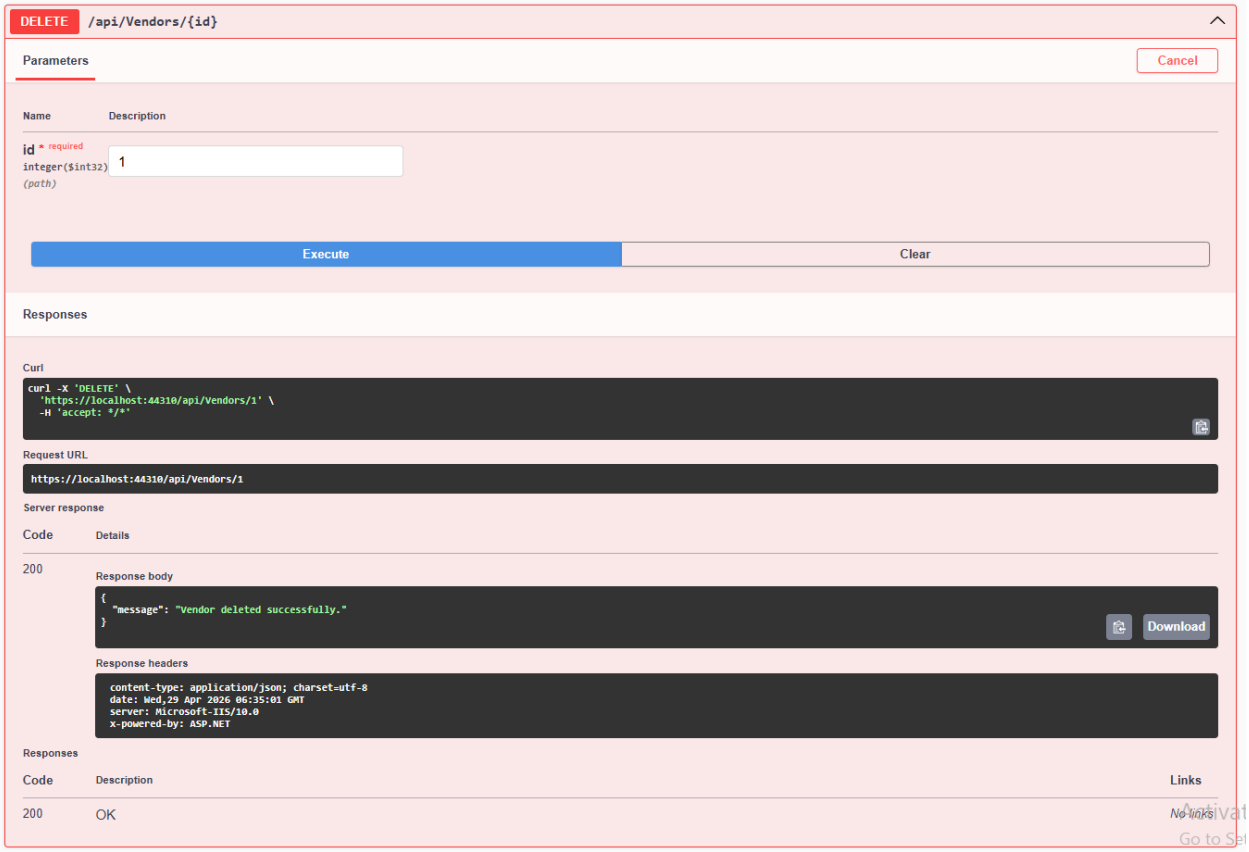


Figure 11: Update Vendor Details



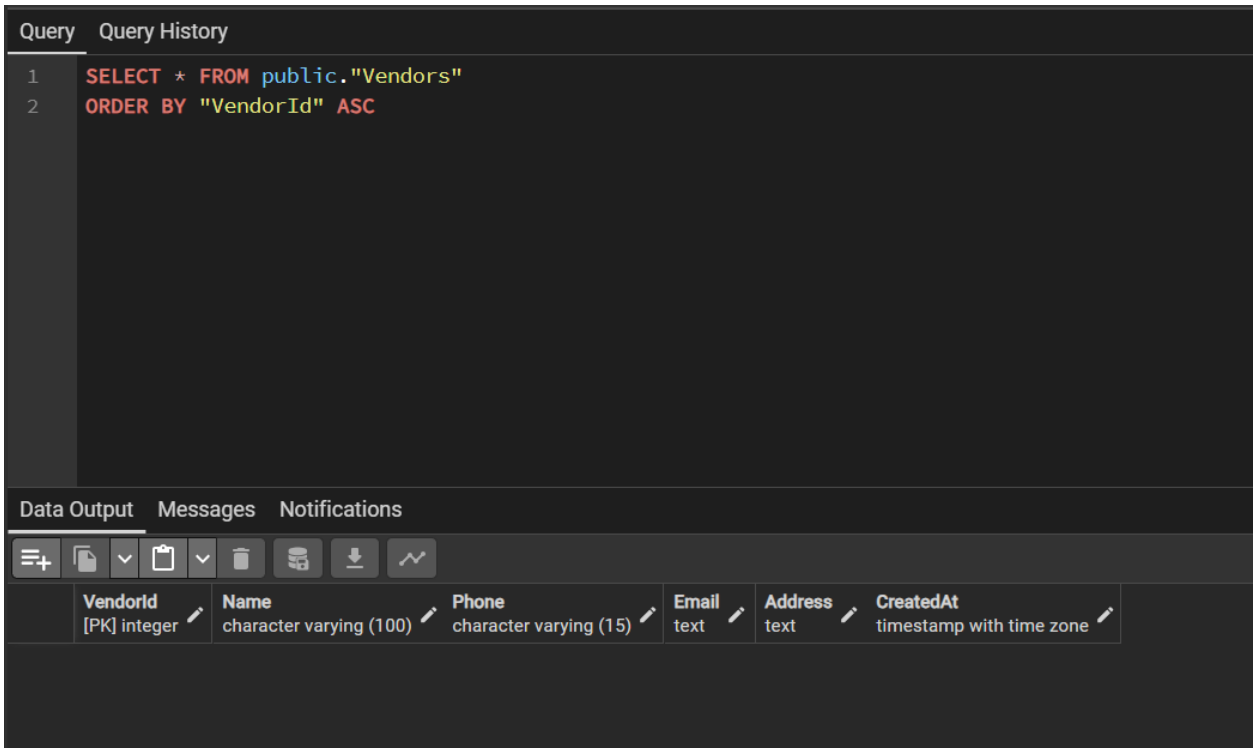


Figure 12: Delete Vendor

3.2.4 Frontend:

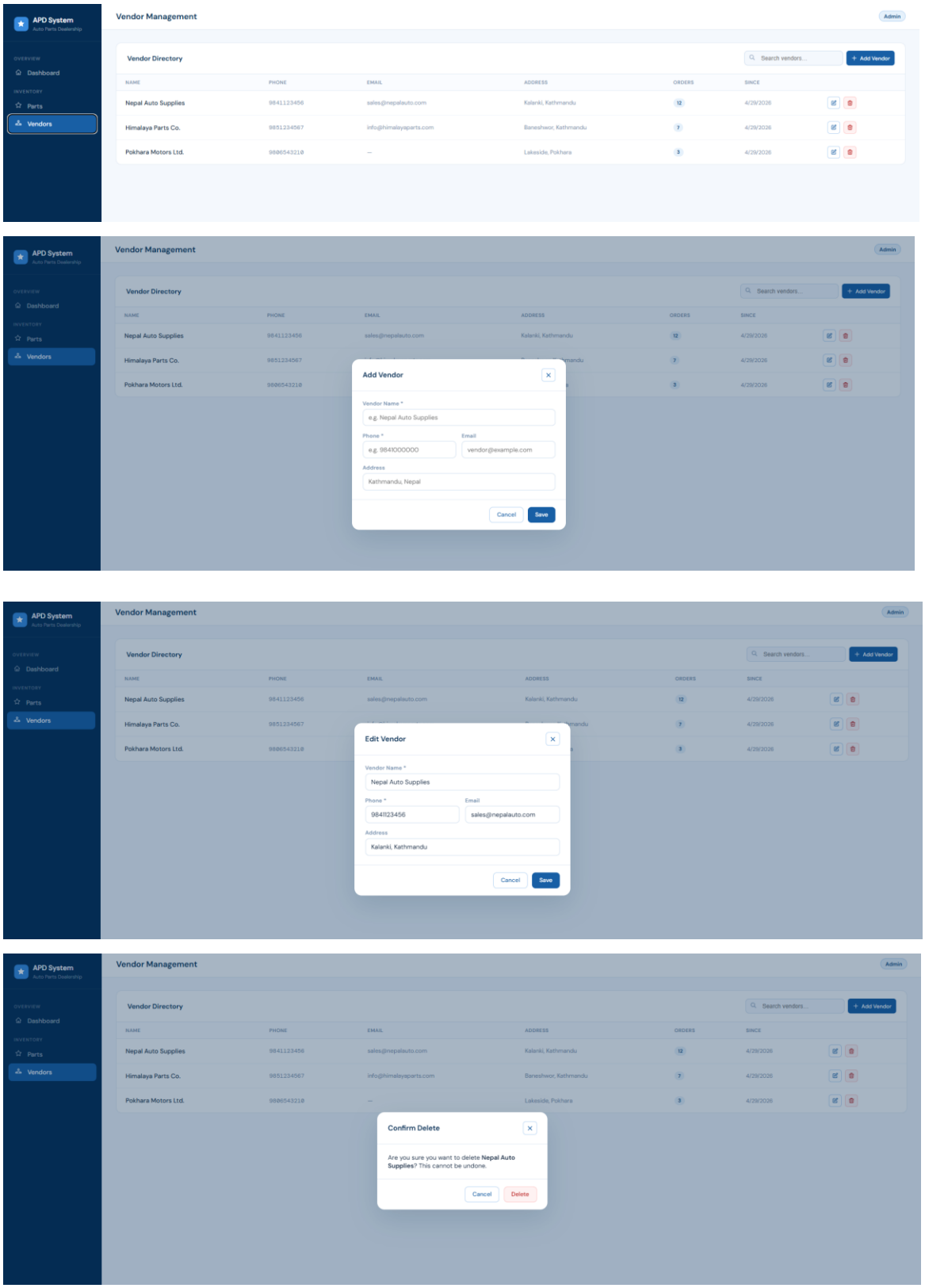


Figure 13: Frontend part of Vendor And Parts management

3.2.5 Member 3: Kushal Chandra Shrestha

Feature 13:

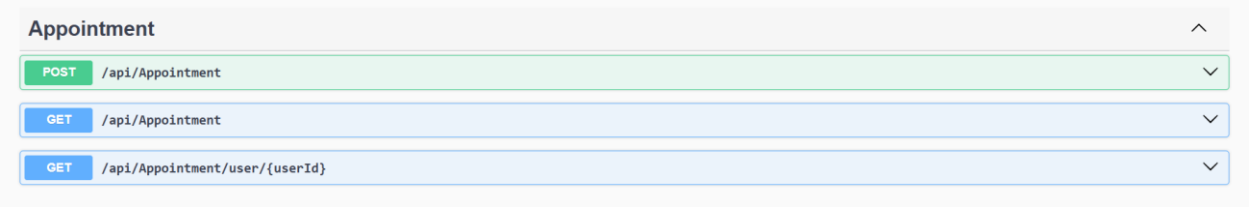


Figure 14: List of API for managing Appointment

- POST /api/appointment - Creates a new appointment for user, takes user id, vehicle id, appointment date time and defaults to pending status.
- GET /api/appointment- Fetches all the appointments for the system.
- GET /api/appointment/user/{userId}- Only fetches the appointment of a particular user.

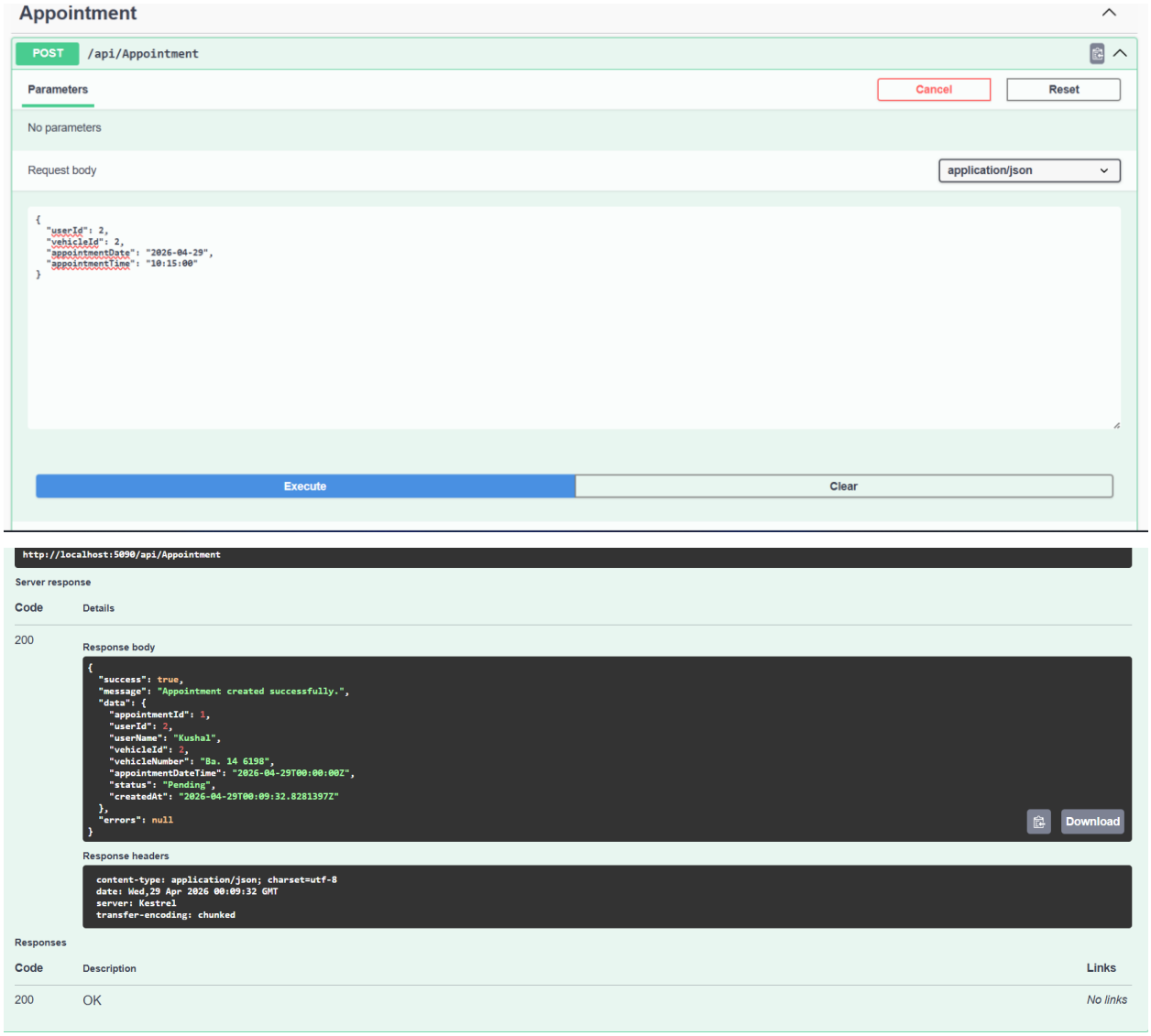


Figure 15: Create Appointment Request

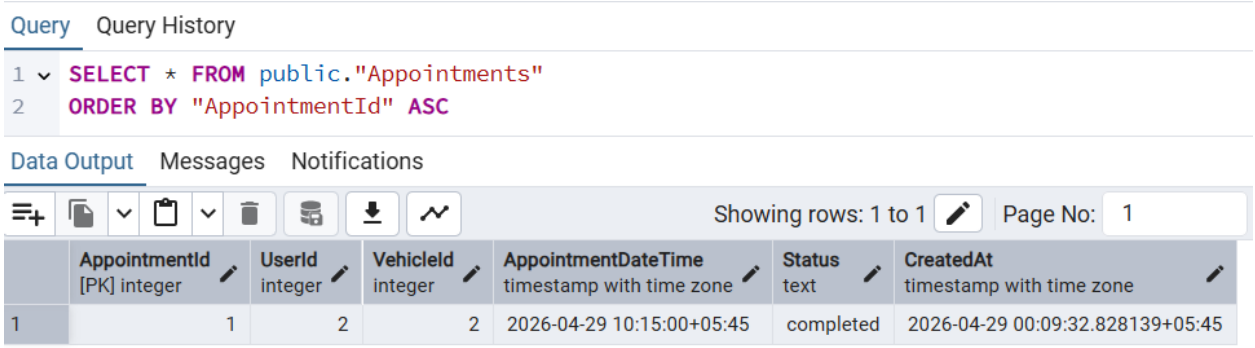


Figure 16: Appointment Saved in Database

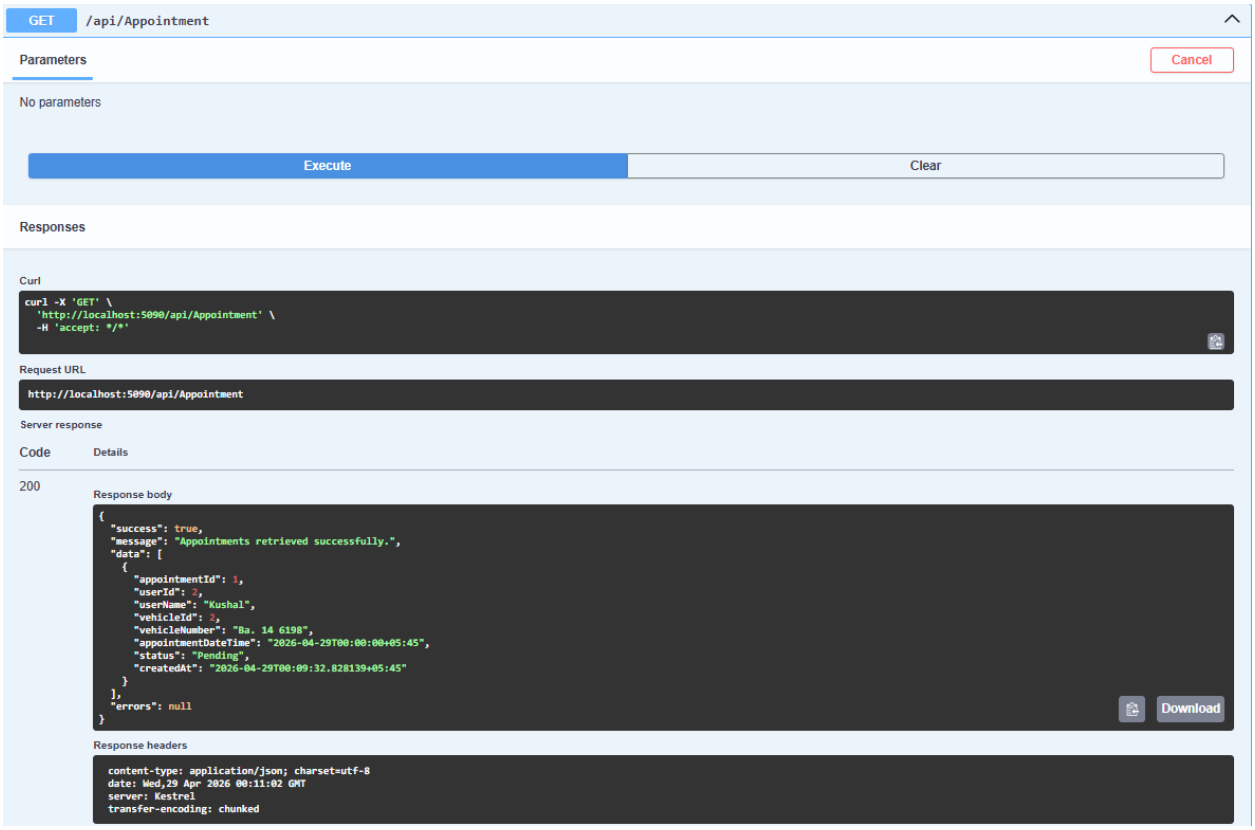


Figure 17: Get All Appointments (GET /api/appointment)

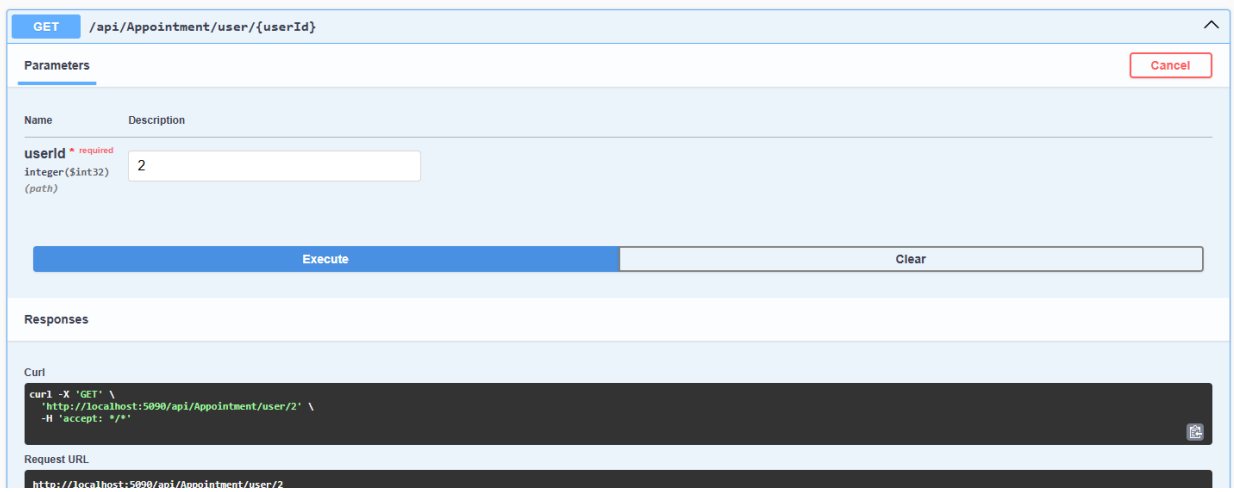


Figure 18: Fetch User Appointments by searching with userId (GET /api/appointment/user/{userId})

Figure 19: Successfully Retrieved Appointment by userId

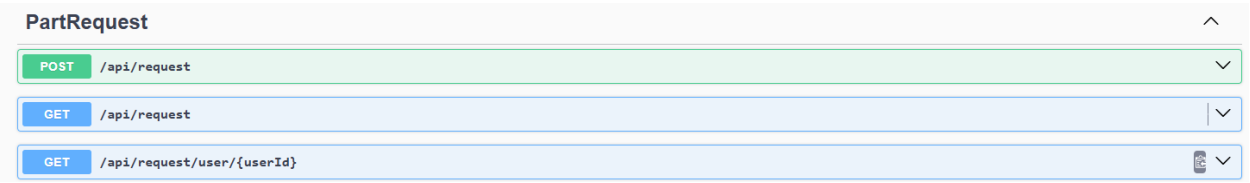


Figure 20: Part request API

- POST /api/request - Creates a part request for unavailable parts.
- GET /api/request - Gets all the part requests in the system.
- GET /api/request/user/{userId}- Gets the part requests for a particular user.

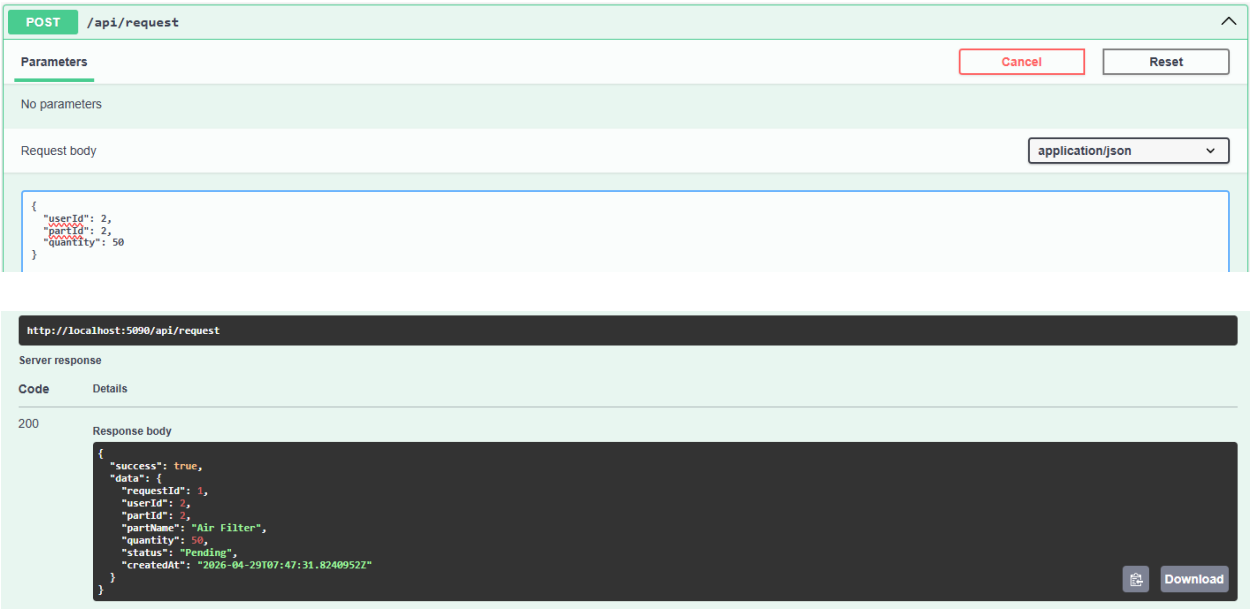


Figure 21: Creates a part request for unavailable parts.

Query | Query History

```
1  SELECT * from "PartRequests"
```

Data Output | Messages | Notifications

Showing row

	RequestId [PK] integer	UserId integer	PartId integer	Quantity integer	Status text	CreatedAt timestamp without time zone
1	1	2	2	50	Pending	2026-04-29 07:47:31.824095

GET /api/request

Parameters | Cancel

No parameters

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:5090/api/request' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:5090/api/request
```

Request URL

```
http://localhost:5090/api/request
```

Server response

Code	Details
200	Response body


```
{
  "success": true,
  "data": [
    {
      "requestId": 2,
      "userId": 3,
      "partId": 4,
      "partName": "Spark Plug (Set of 4)",
      "quantity": 50,
      "status": "Pending",
      "createdAt": "2026-04-29T07:48:41.569082Z"
    },
    {
      "requestId": 1,
      "userId": 2,
      "partId": 2,
      "partName": "Air Filter",
      "quantity": 50,
      "status": "Pending",
      "createdAt": "2026-04-29T07:47:31.824095Z"
    }
  ]
}
```


Download

Response headers

```
content-type: application/json; charset=utf-8
date: Wed,29 Apr 2026 07:48:48 GMT
server: Kestrel
transfer-encoding: chunked
```

Responses

Code	Description	Links
200	Success	*No links*

Figure 22: Get All Part Requests

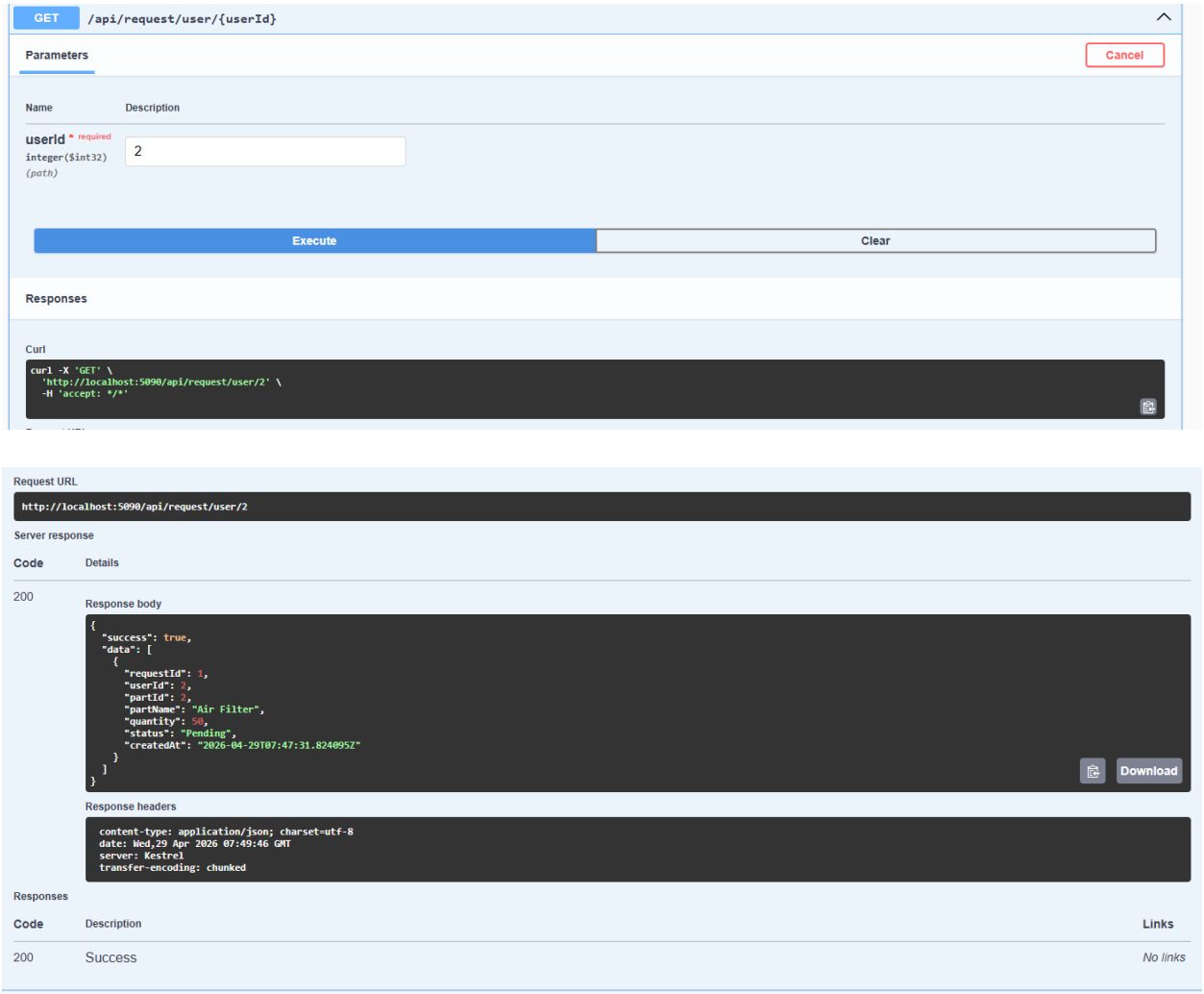


Figure 23: Get User Part Requests

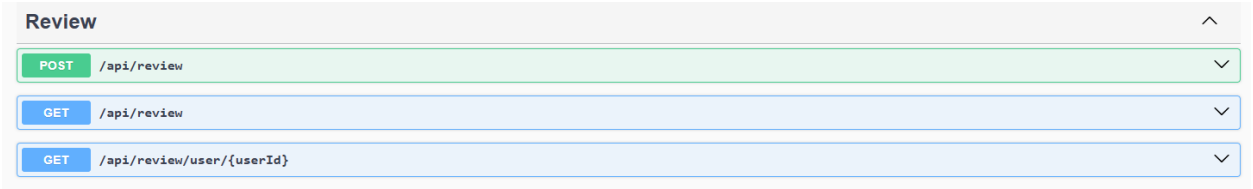
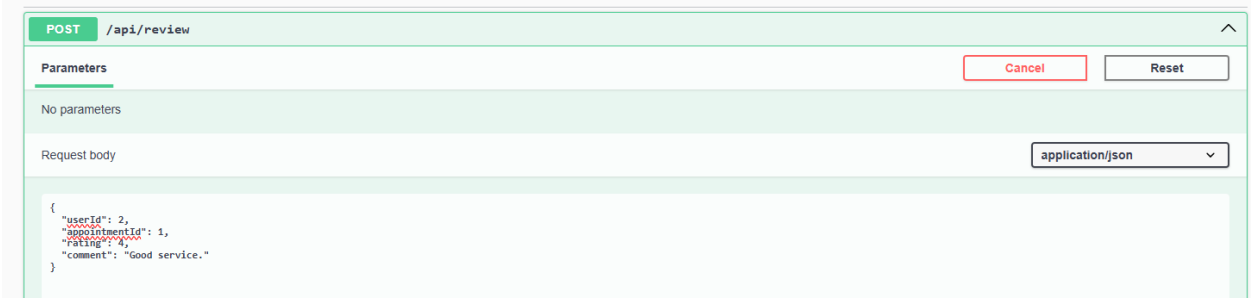
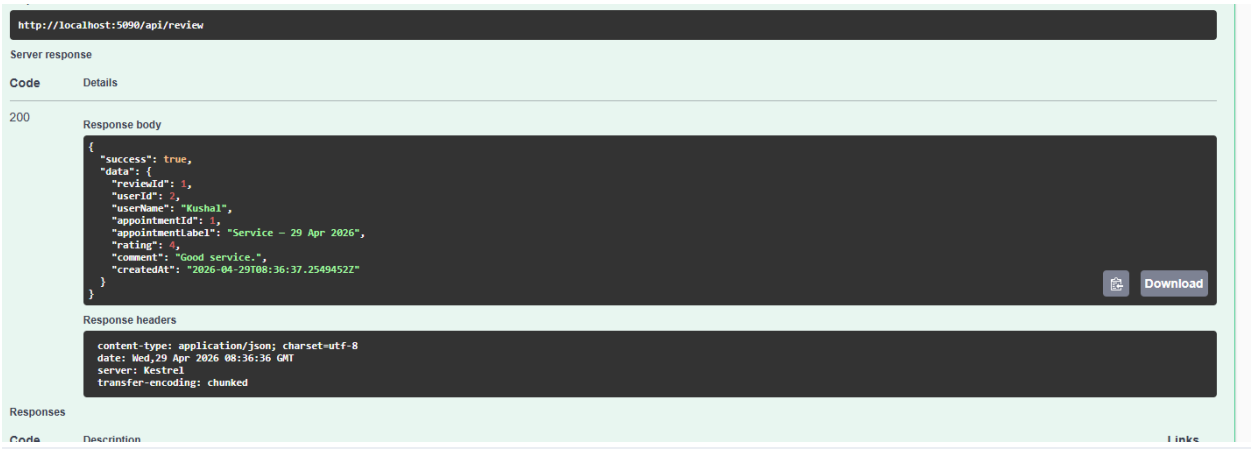
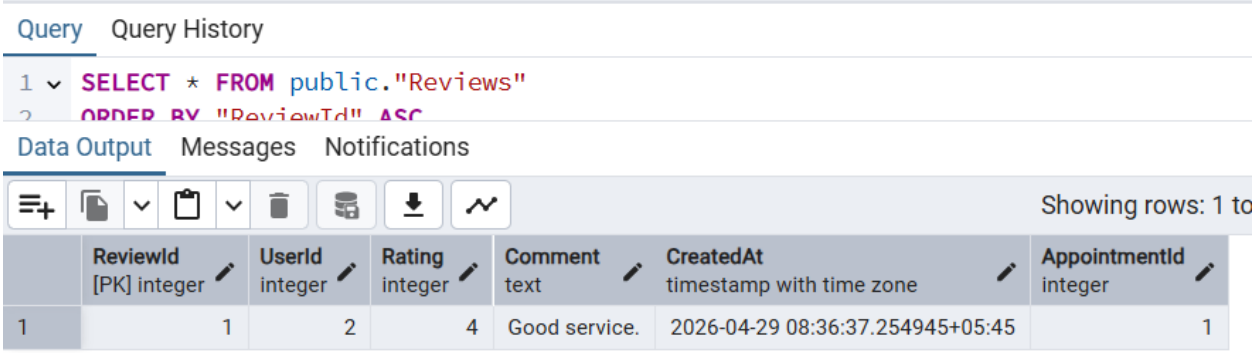


Figure 24: List of Api to POST and GET the review

- POST /api/review- Creates a review for an appointment for user.
- GET /api/review- Fetch all the reviews in the system.
- GET /api/reviews/user/{userId}- Fetches only the reviews of one user .







	ReviewId [PK] integer	UserId integer	Rating integer	Comment text	CreatedAt timestamp with time zone	AppointmentId integer
1	1	2	4	Good service.	2026-04-29 08:36:37.254945+05:45	1

GET /api/review

Parameters Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:5090/api/review' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:5090/api/review
```

Server response

Code Details

200

Response body


```
{
  "success": true,
  "data": [
    {
      "reviewId": 1,
      "userId": 2,
      "userName": "Kushal",
      "appointmentId": 1,
      "appointmentLabel": "Service – 29 Apr 2026",
      "rating": 4,
      "comment": "Good service.",
      "createdAt": "2026-04-29T08:36:37.254945+05:45"
    }
  ]
}
```


Download

GET /api/review/user/{userId}

Parameters Cancel

Name	Description
userId * required integer($int32) (path)	2

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:5090/api/review/user/2' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:5090/api/review/user/2
```

Server response

Code Details

200

Response body


```
{
  "success": true,
  "data": [
    {
      "reviewId": 1,
      "userId": 2,
      "userName": "Kushal",
      "appointmentId": 1,
      "appointmentLabel": "Service – 29 Apr 2026",
      "rating": 4,
      "comment": "Good service.",
      "createdAt": "2026-04-29T08:36:37.254945+05:45"
    }
  ]
}
```

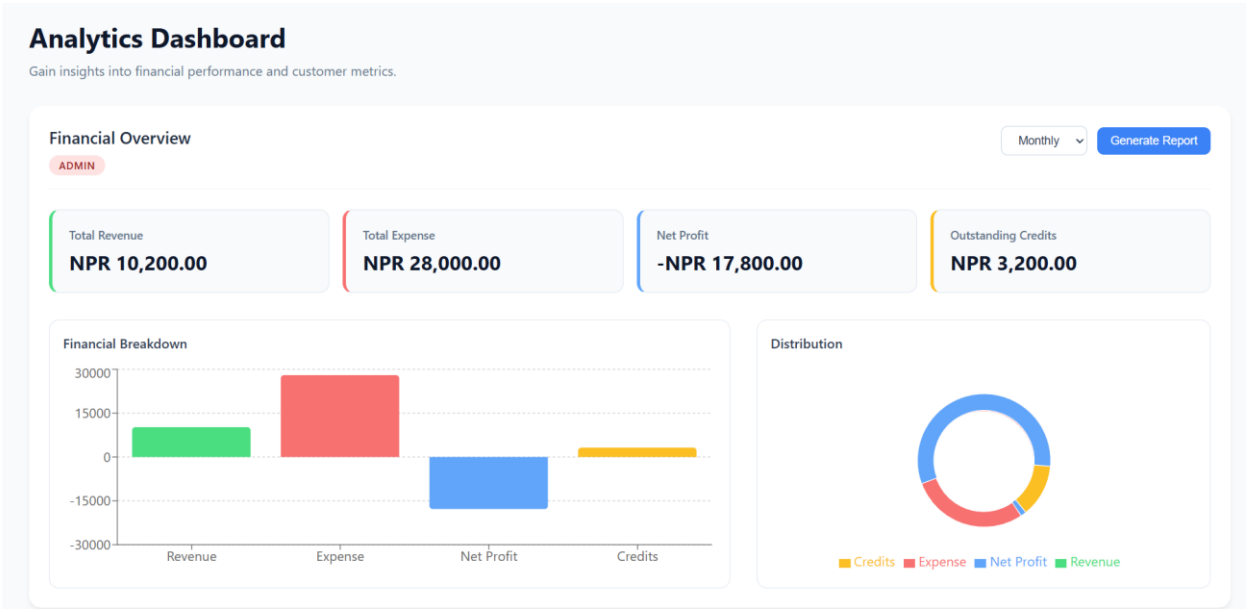

Download

Response headers

3.2.6 Member 4: Pratik Prasai

Feature 1: Admin can generate and view financial reports (daily, monthly, yearly)

Frontend:



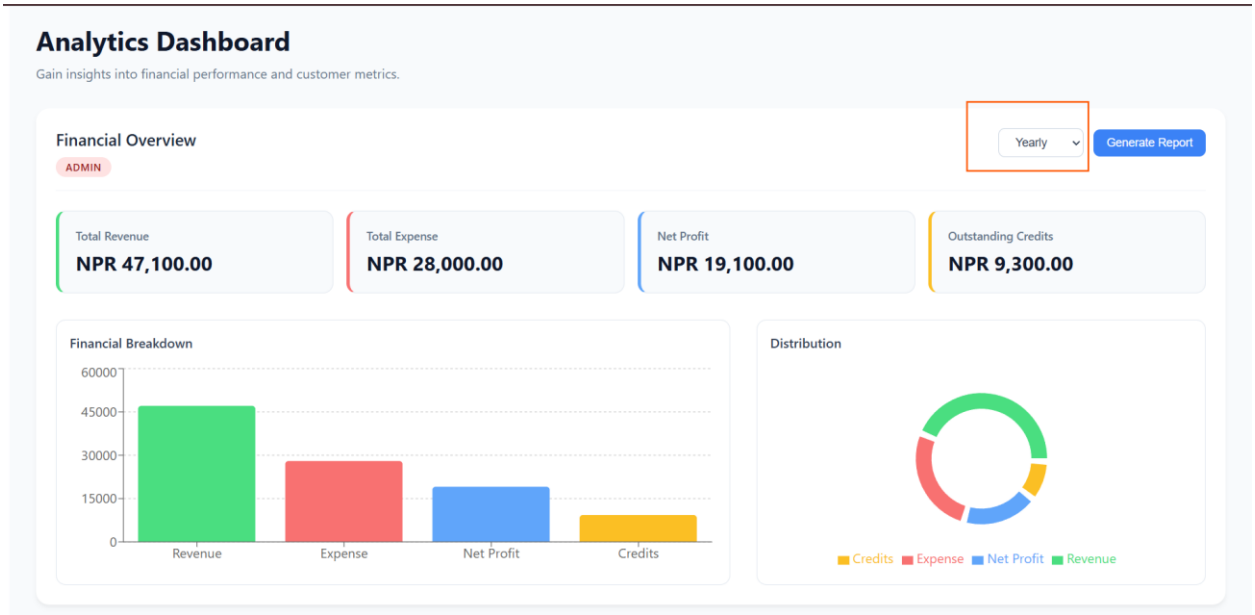
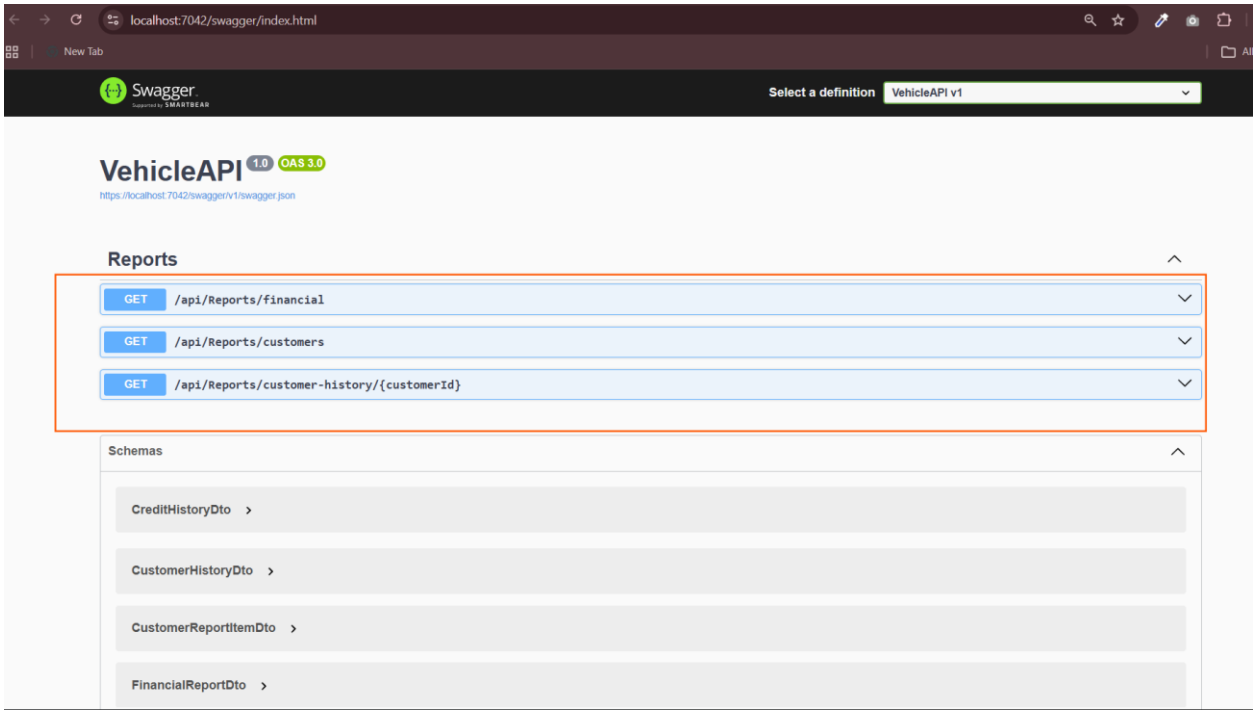
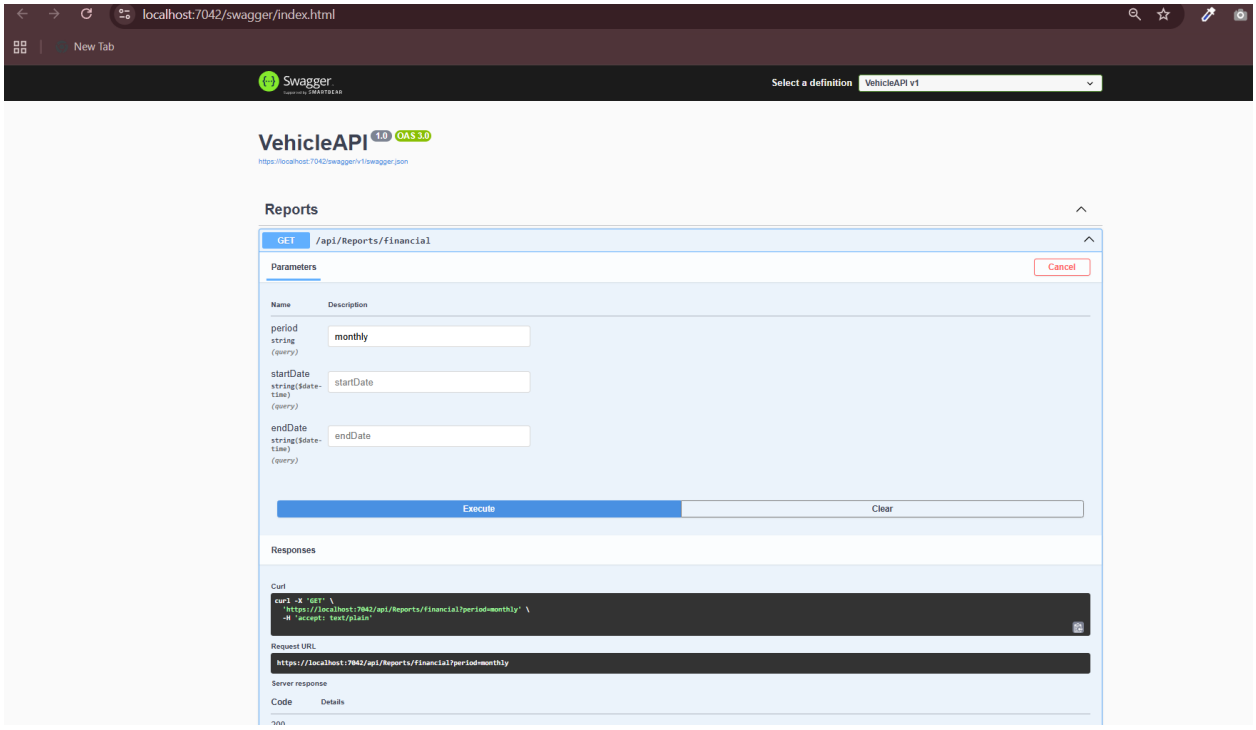


Figure 25: Yearly Financial Report

Backend:


localhost:7042/swagger/index.html
New Tab
Swagger
Select a definition
VehicleAPI v1
VehicleAPI
1.0
OAS 3.0
https://localhost:7042/swagger/v1/swagger.json
Reports
GET /api/Reports/financial
GET /api/Reports/customers
GET /api/Reports/customer-history/{customerId}
Schemas
CreditHistoryDto
CustomerHistoryDto
CustomerReportItemDto
FinancialReportDto


localhost:7042/swagger/index.html
New Tab
Swagger
Select a definition
VehicleAPI v1
VehicleAPI
1.0
OAS 3.0
Reports
GET /api/Reports/financial
Parameters
Cancel
Name
Description
period
string
(query)
monthly
startDate
string($date-time)
(query)
startDate
endDate
string($date-time)
(query)
endDate
Execute
Clear
Responses
Curl
curl -X 'GET' \
'https://localhost:7042/api/Reports/financial?period=monthly' \
-H 'accept: text/plain'
Request URL
https://localhost:7042/api/Reports/financial?period=monthly
Server response
Code
Details

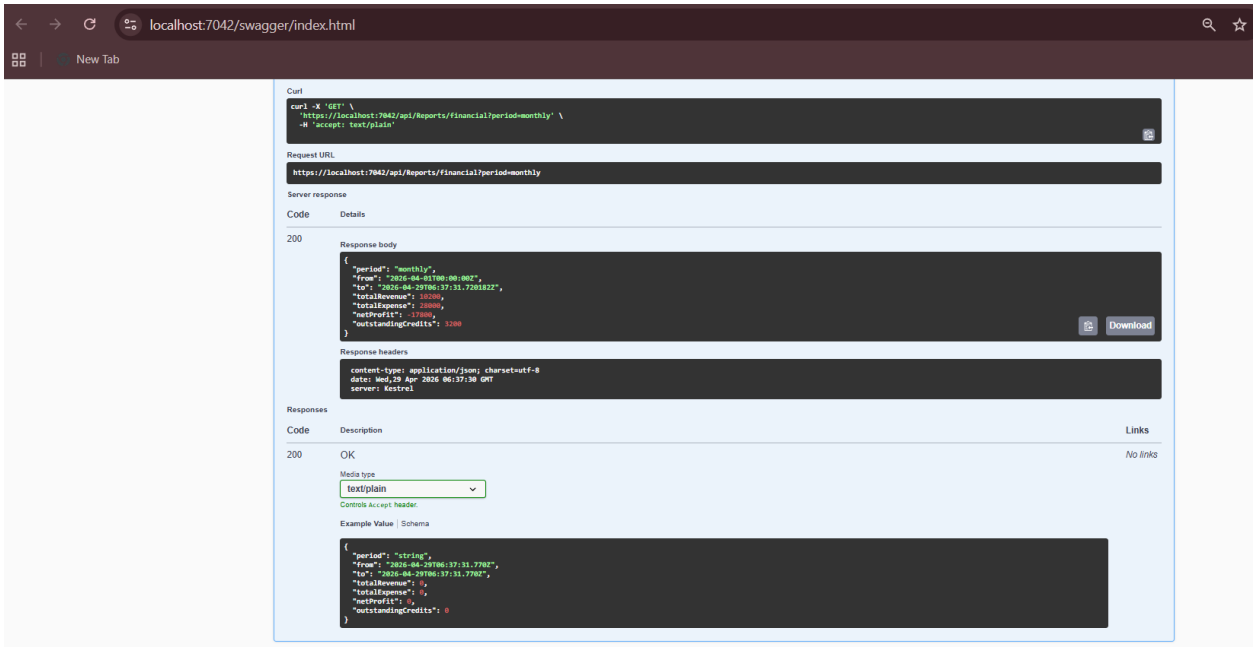
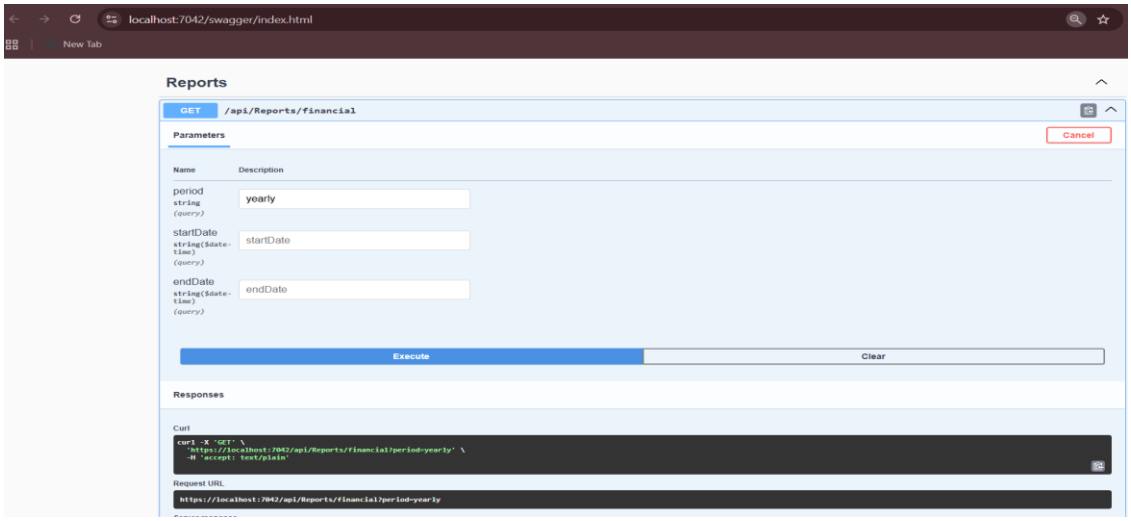


Figure 26: Api to fetch monthly financial report



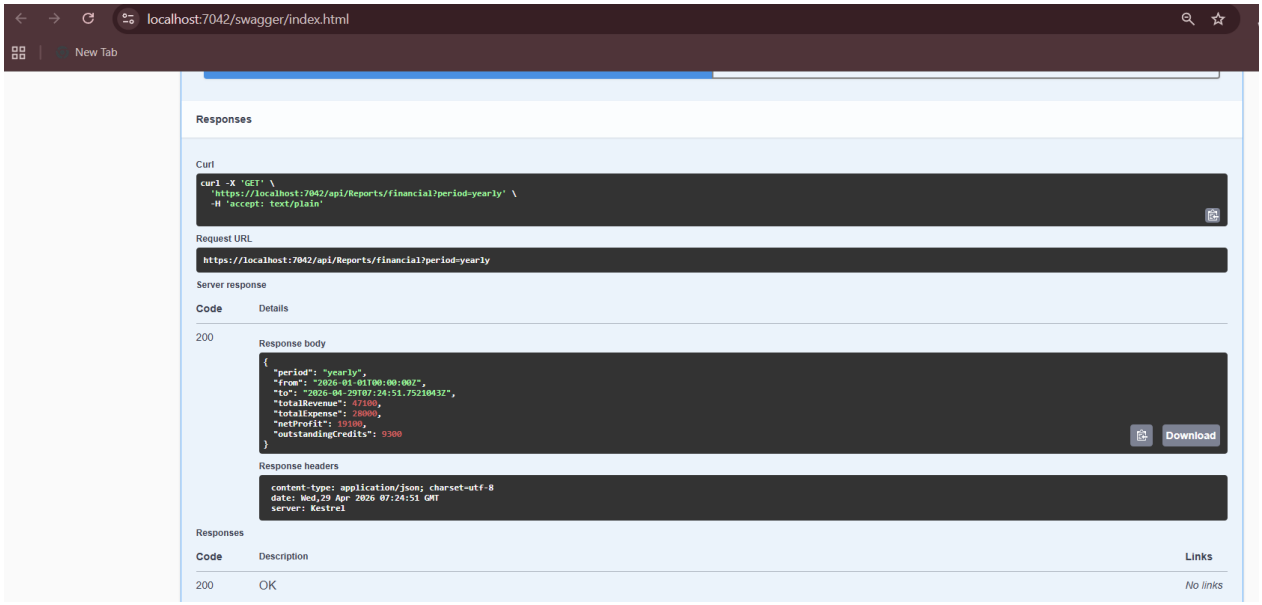
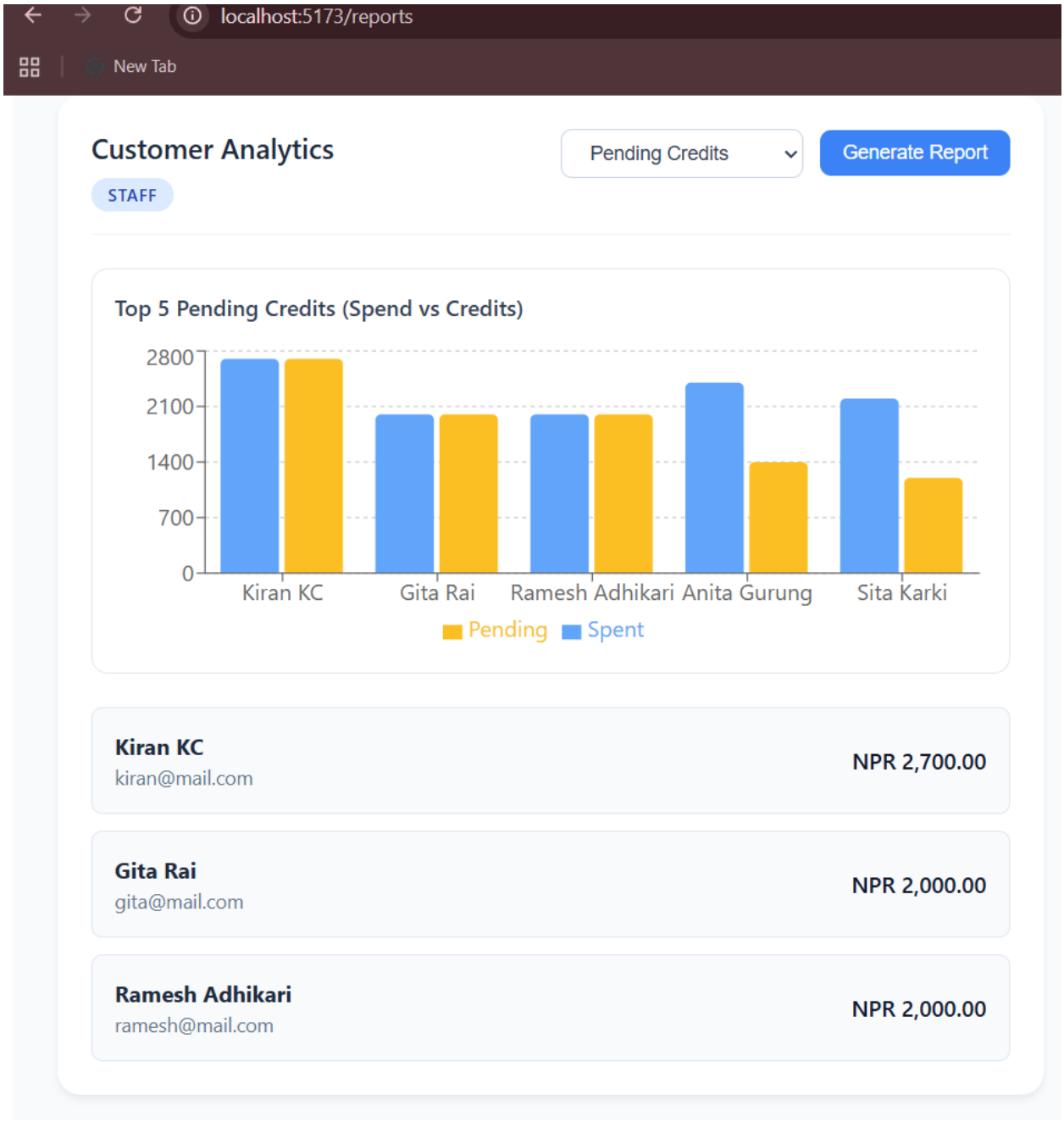


Figure 27: Financial Report Yearly

GET /api/reports/financial returns a financial analytics summary for a selected time range. It totals revenue by summing Sales.FinalAmount, totals expense by summing Purchases.TotalAmount, calculates net profit as revenue minus expense, and also totals outstanding unpaid credits by summing Credits.AmountDue where IsPaid is false. The time window is controlled by the period query parameter (daily, monthly default, or yearly), and you can override that by providing startDate and endDate (both optional).

Feature 9: Staff can generate customer-related reports (regulars, high spenders, pending credits)

Frontend:



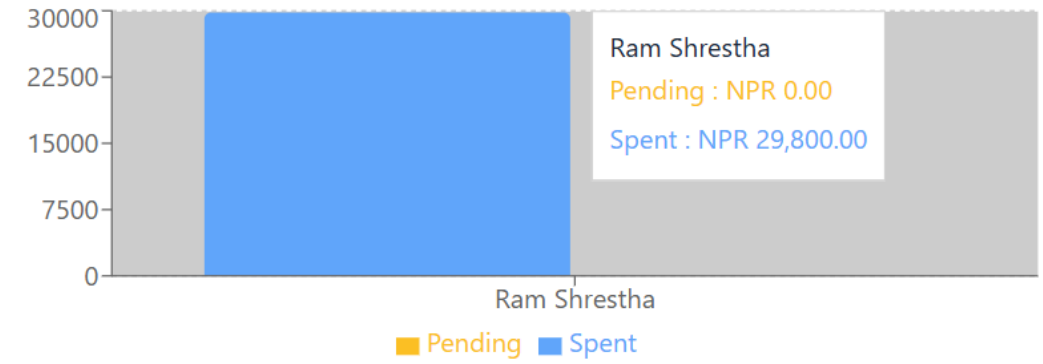
Customer Analytics

STAFF

High Spenders

Generate Report

Top 5 High Spenders (Spend vs Credits)



Ram Shrestha
ram@mail.com

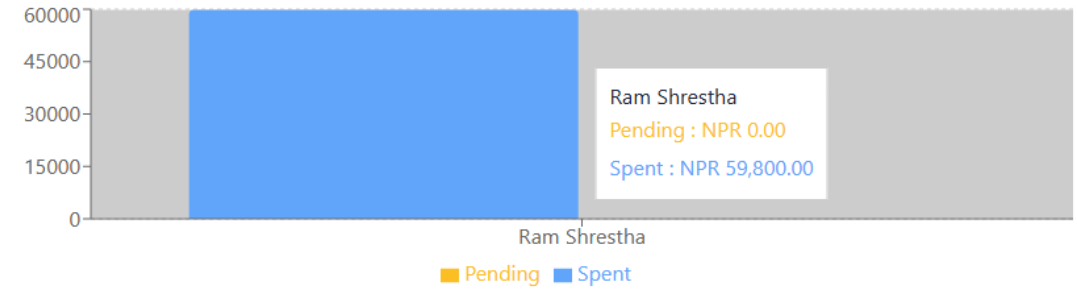
NPR 29,800.00

Customer Analytics

STAFF

Regular Customers | Generate Report

Top 5 Regular Customers (Spend vs Credits)



Ram Shrestha
ram@mail.com

NPR 59,800.00

Backend:

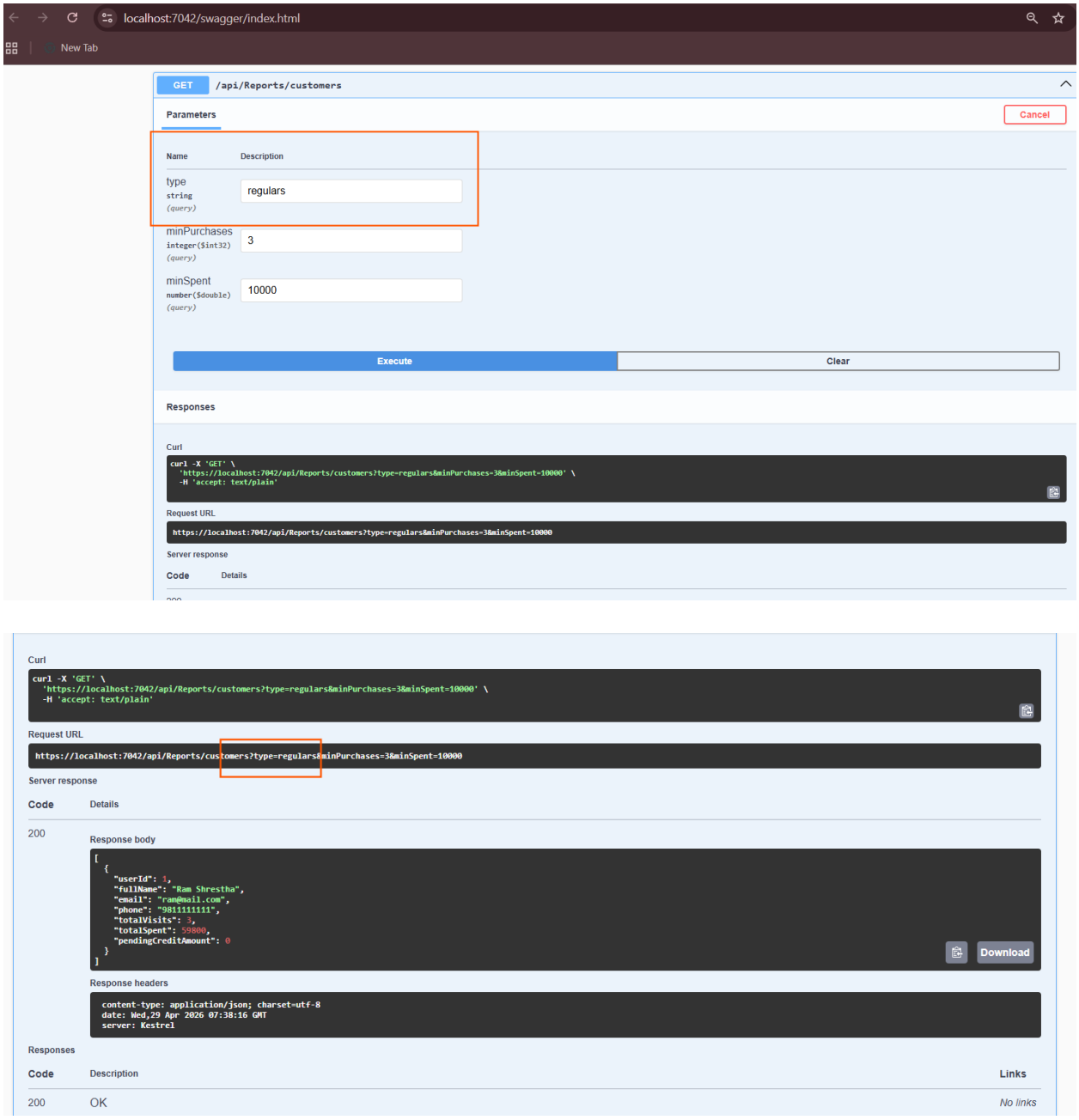
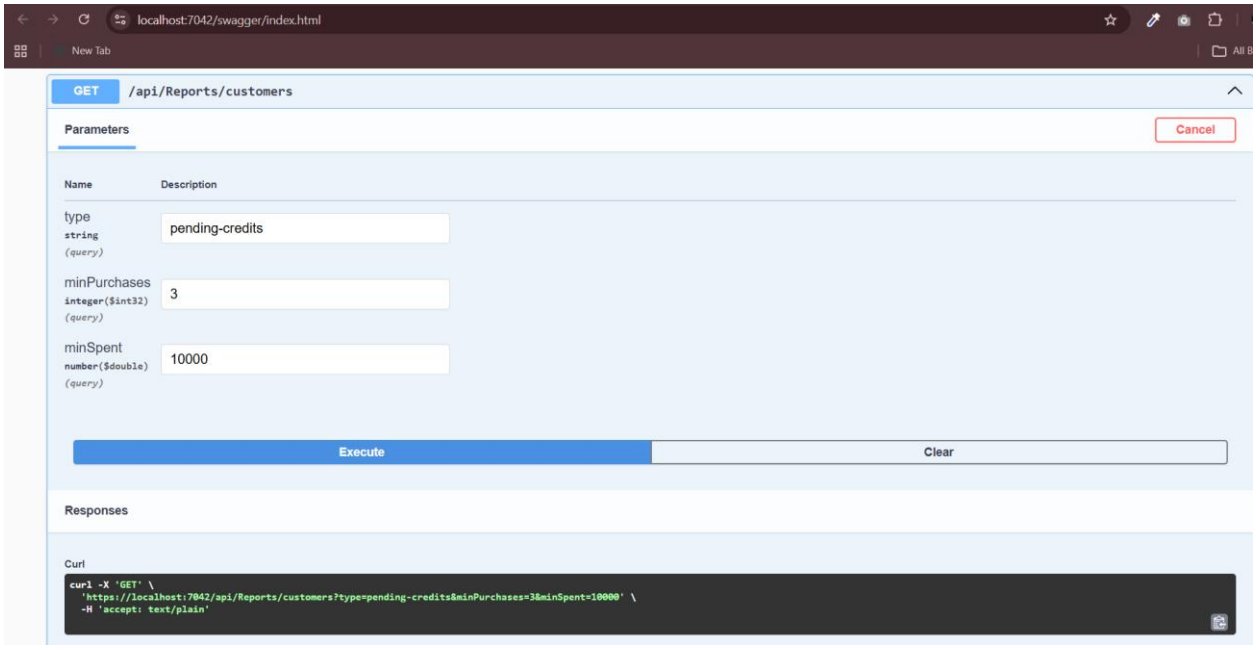


Figure 28: Regulars Report Response



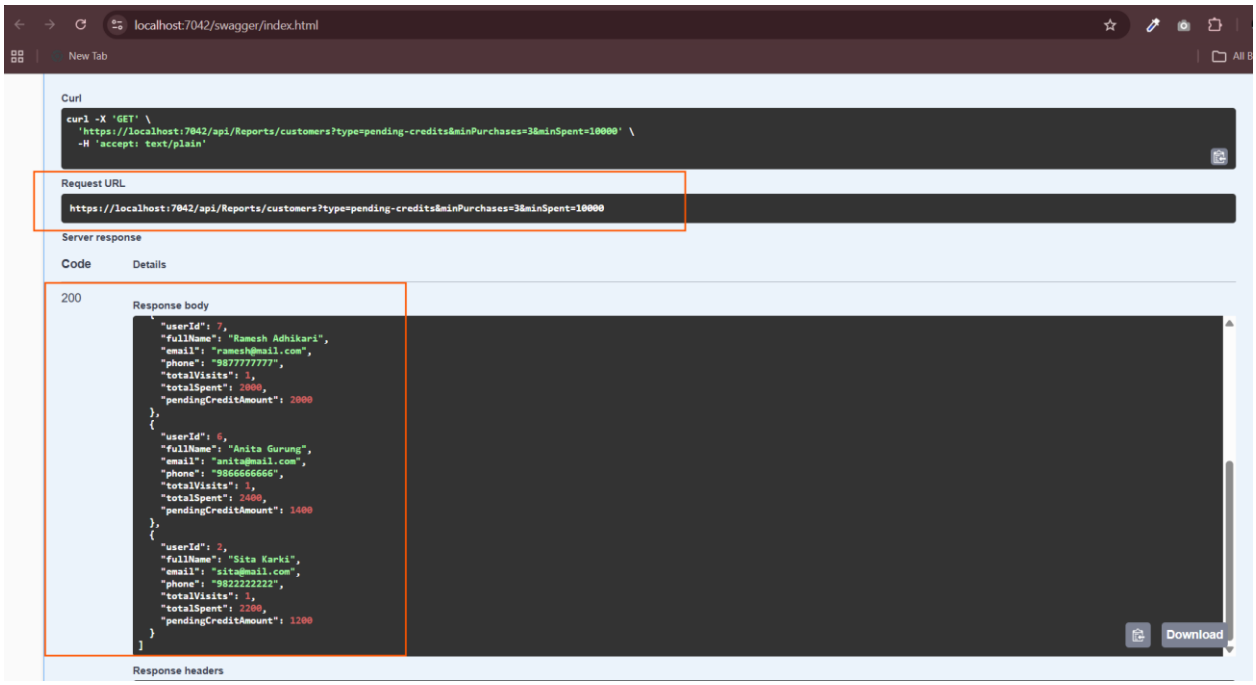


Figure 29: Pending Credits Report Response

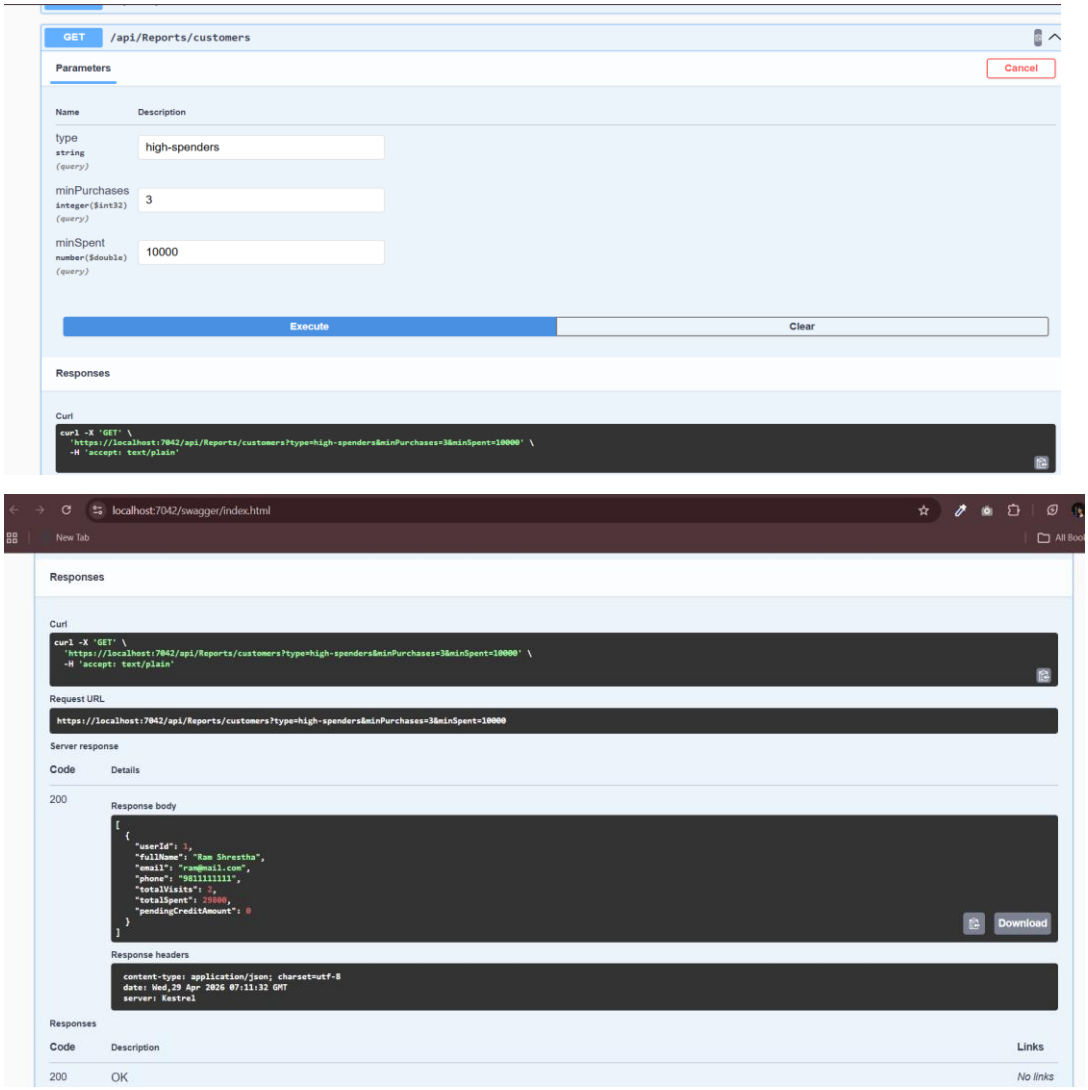


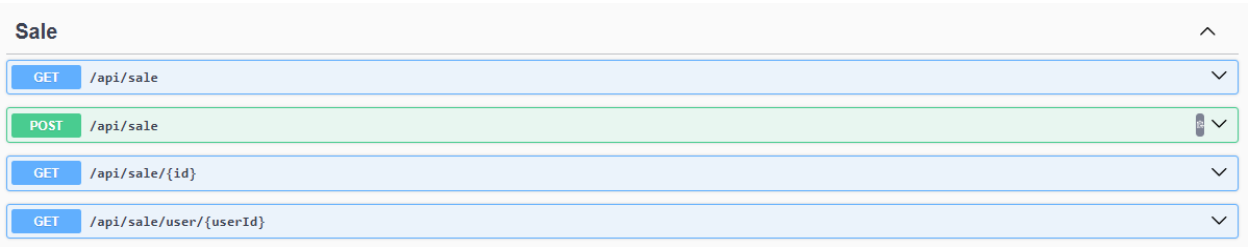
Figure 30: High spender's Report Response

The /api/reports/customers API provides a simple report about customers (specifically users with RoleId = 3). For each customer, it calculates how many times they have made purchases (total visits), how much money they have spent in total (total spent), and how much money they still owe (pending credit). The API also allows filtering results using a type parameter. For example, you can get regular customers who have made a minimum number of purchases (default is 3), high-spending customers who have spent above a certain amount (default is 10,000), or customers who still have unpaid credit. In the end, it returns a list of customers along with these calculated details.

3.2.7 Member 5: Slesha Rawal

Feature 7: Staff can sell vehicle parts and create sales invoices

Backend:



- GET /api/sale: Fetches the whole list of Sales records from the whole system.
- POST /api/sale: Creates the whole new sale order.
- GET /api/sale/{id}: Fetches the receipt/invoice details of individual sale using Sale ID.
- GET /api/sale/user/{userId}: Fetches the history of sale of individual customers.

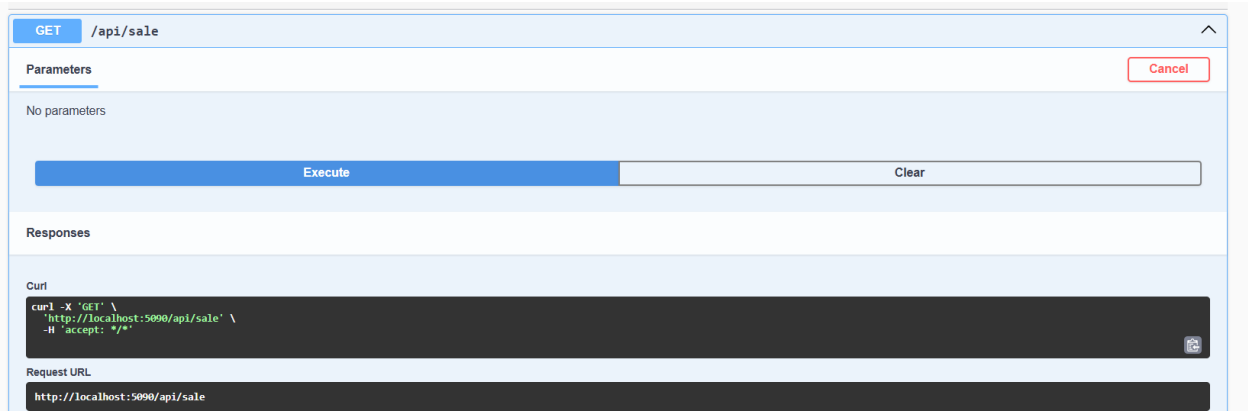


Figure 31: Fetch all the available sales

200

Response body


```
{
  "success": true,
  "data": [
    {
      "saleId": 2,
      "userId": 5,
      "userName": "Ram Sharma",
      "totalAmount": 11200,
      "discount": 0,
      "finalAmount": 11200,
      "amountPaid": 0,
      "paymentStatus": "Unpaid",
      "createdAt": "2026-04-29T08:47:30.378935+05:45",
      "items": [
        {
          "saleItemId": 3,
          "partId": 8,
          "partName": "Wiper Blade Set",
          "quantity": 20,
          "unitPrice": 450,
          "subtotal": 9000
        },
        {
          "saleItemId": 4,
          "partId": 6,
          "partName": "Radiator Coolant 1L",
          "quantity": 10,
          "unitPrice": 220,
          "subtotal": 2200
```


Download

Response headers

```
content-type: application/json; charset=utf-8
date: Wed,29 Apr 2026 08:48:27 GMT
server: Kestrel
transfer-encoding: chunked
```

POST /api/sale

Parameters

Cancel Reset

No parameters

Request body application/json


```
{
  "userId": 5,
  "items": [
    {
      "partId": 8,
      "quantity": 20
    },
    {
      "partId":6,
      "quantity":10
    }

  ],
  "discount": 0,
  "amountPaid": 0
}
```


Request URL

```
http://localhost:5090/api/sale
```

Server response

Code Details

200

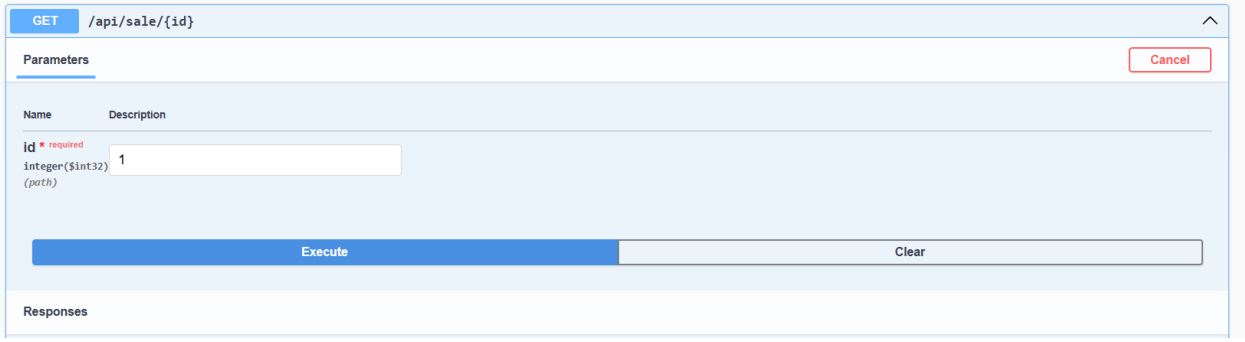
Response body


```
{
  "success": true,
  "data": {
    "saleId": 2,
    "userId": 5,
    "userName": "Ram Sharma",
    "totalAmount": 11200,
    "discount": 0,
    "finalAmount": 11200,
    "amountPaid": 0,
    "paymentStatus": "Unpaid",
    "createdAt": "2026-04-29T08:47:30.3789353Z",
    "items": [
      {
        "saleItemId": 3,
        "partId": 8,
        "partName": "Wiper Blade Set",
        "quantity": 20,
        "unitPrice": 450,
        "subtotal": 9000
      },
      {
        "saleItemId": 4,
        "partId": 6,
        "partName": "Radiator Coolant 1L",
        "quantity": 10,
        "unitPrice": 220,
        "subtotal": 2200
      }
```


Download

Response headers

Figure 32: Create Sale Request


GET
/api/sale/{id}
Parameters
Cancel
Name
Description
id * required
integer($int32)
(path)
1
Execute
Clear
Responses


Code
Details
200
Response body
{
"success": true,
"data": {
"saleId": 1,
"userId": 2,
"userName": "Kushal",
"totalAmount": 26200,
"discount": 0,
"finalAmount": 26200,
"amountPaid": 0,
"paymentStatus": "Unpaid",
"createdAt": "2026-04-29T08:46:53.255311+05:45",
"items": [
{
"saleItemId": 1,
"partId": 5,
"partName": "Timing Belt",
"quantity": 20,
"unitPrice": 1200,
"subtotal": 24000
},
{
"saleItemId": 2,
"partId": 6,
"partName": "Radiator Coolant 1L",
"quantity": 10,
"unitPrice": 220,
"subtotal": 2200
}
Down
Response headers
content-type: application/json; charset=utf-8
date: Wed,29 Apr 2026 08:49:06 GMT
server: Kestrel
transfer-encoding: chunked
Responses
Code
Description

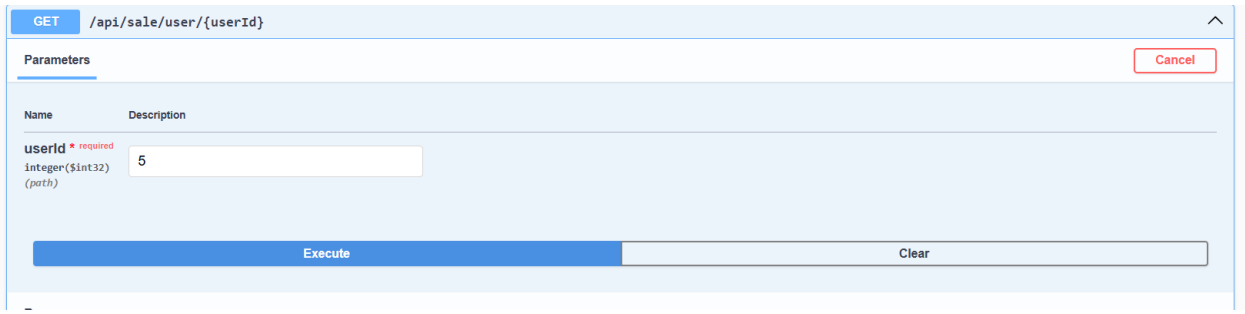
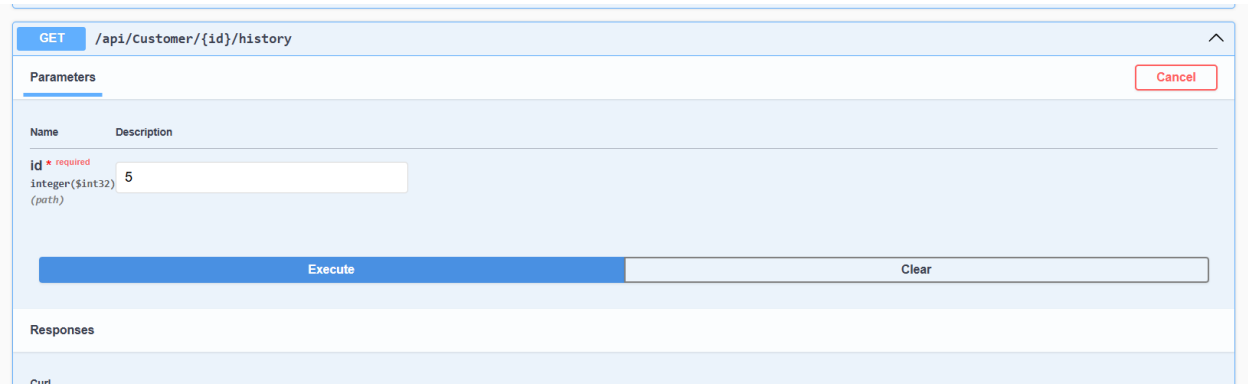

GET
/api/sale/user/{userId}
Parameters
Cancel
Name
Description
userId * required
integer($int32)
(path)
5
Execute
Clear

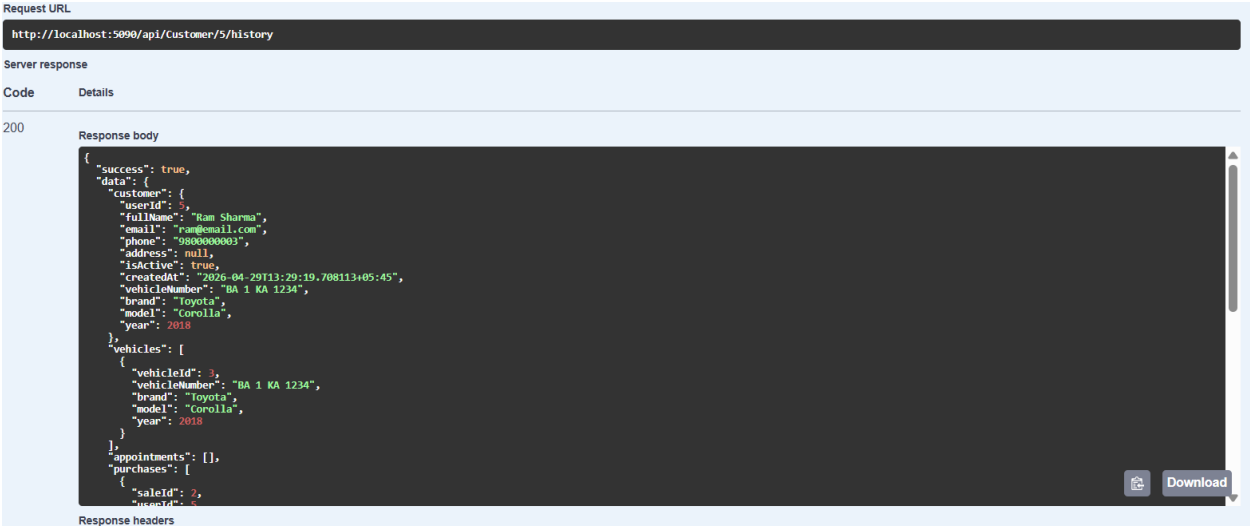
Figure 33: Generated invoice of sales

Feature 8: Staff can view customer details, history, and vehicle info

Backend:



- GET /api/customer/{id}/history- fetches the customer history like appointment , service and vehicle details.



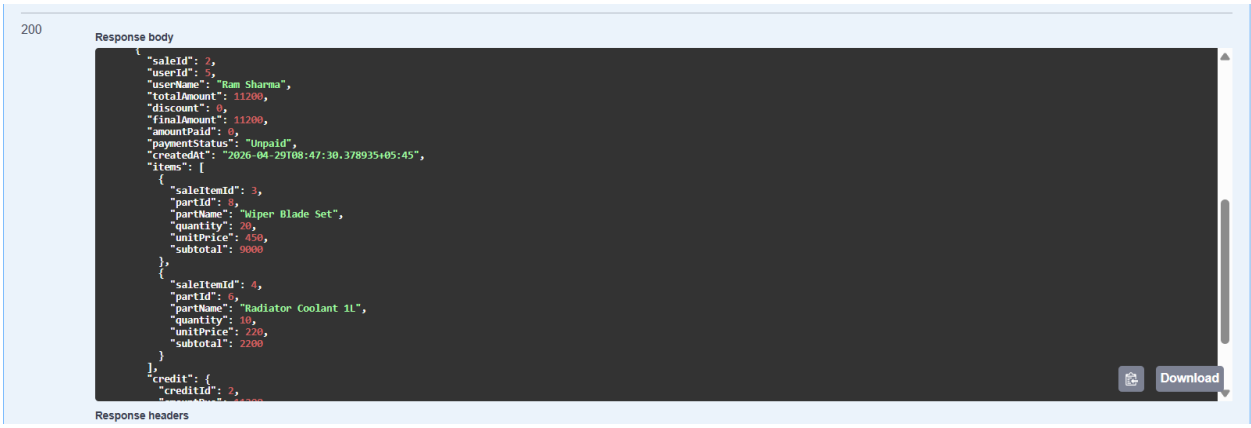


Figure 34: Customer Purchase History Retrieval

4 Proof of Work

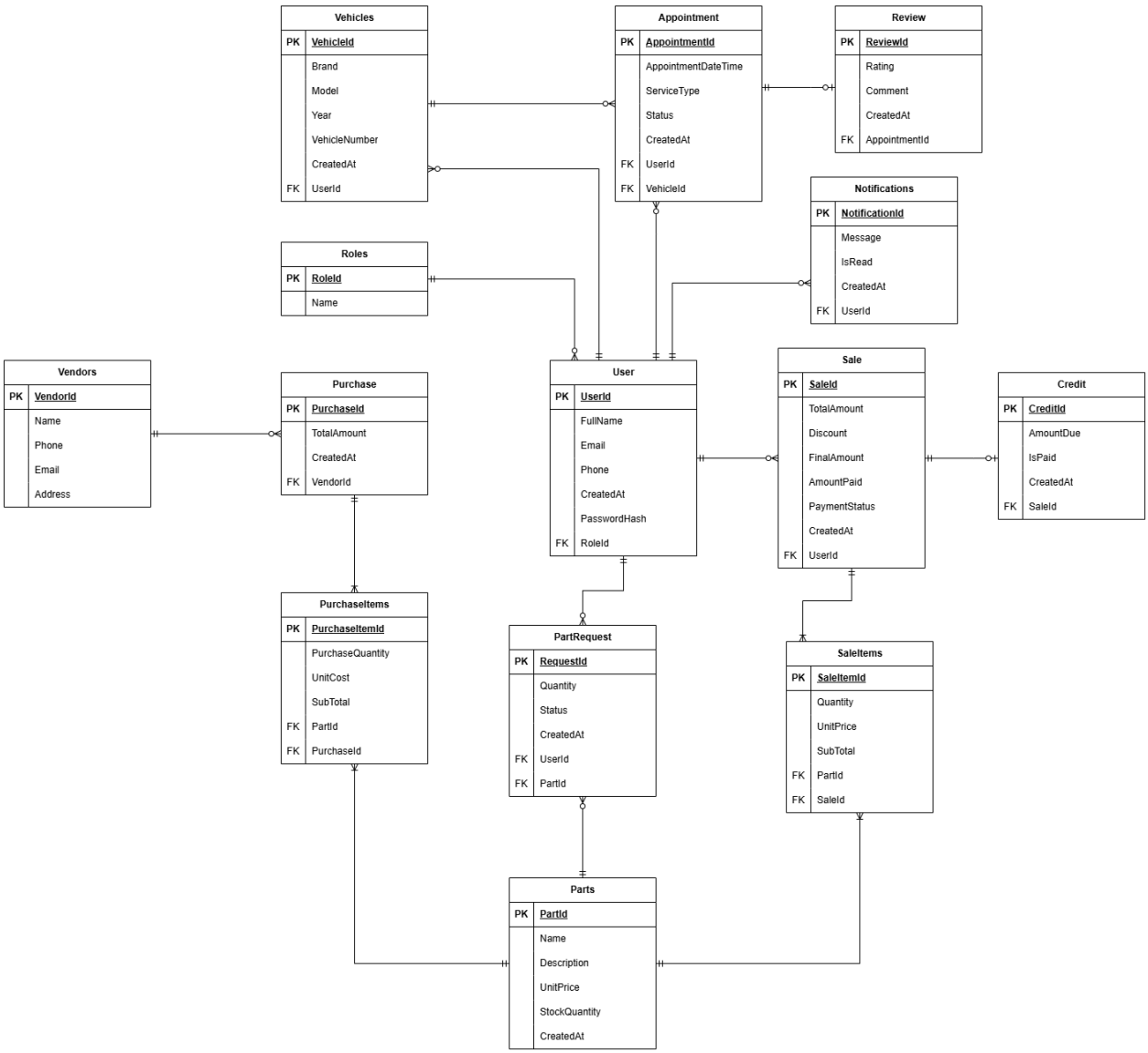


Figure 35: ER Diagram Representing Tables and Their Relationships

5 Individual reflection

5.1 Member 1: Bipin Kumar Thapa

I focused on customer and admin management. So far, all the customer registration and its associated vehicles, customer searching (by vehicle number, phone, ID, name) and staff registration and handling of their roles (admin panel planning/structure) are completed. In this regard, the task has enhanced the management of customer and staff information in the system.

Some pending tasks are admin staff management (fully handling registration and their roles in admin panel) and customer self registration with their profile and vehicle. When the task is completed, it will complete customer and staff data handling.

5.2 Member 2: Dishant Panjiyar

My responsibility was on managing inventory and supplier. He has already completed the management of parts (purchasing, editing, deleting parts) and suppliers (CRUD operation). He has done this by managing inventory and suppliers and so this functionality has been handled.

His remaining tasks is to manage admin purchase invoice creating process (so that it automatically updates inventory). When he completes this functionality, the inventory data management and tracking will be greatly eased.

5.3 Member 3: Kushal Chandra Shrestha

I covered the functionality that includes all those which interact with the customers (for automated processes). Customer appointment booking, service requests when any part is not available, service review, loyalty program (giving 10% discount if purchase amount is > 5000) has already been developed. Through this functionality, he has established the user's access and interaction to different systems.

The pending tasks are automatic system notification system (low stock alerts for admin, credit reminders to customers having unpaid credit for more than 1 month).

5.4 Member 4 : Pratik Prasai

I covered the reporting system. Financial reports (daily, monthly, yearly) and customer reports (regular customer, high spenders, pending credits) are already completed. These reports have been greatly useful for the management system.

His pending tasks include showing customer purchase history and service history. Through this, customer has his information in his account and admin also know what the customer purchases and which services he got.

5.5 Member 5 : Slesha Rawal

I covered the system of sales and giving details of the customer to staff. The selling process of parts and its Invoice generation and customer details and its history, vehicles can be accessed is already working well. This helps the admin and staff in day to day operations.

The pending tasks are customer email invoicing system. This task will complete the entire process of selling parts.